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Material Characterization using Graph Neural Networks

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Abstract

Material Characterization using Graph Neural Networks

This thesis explores the application of Graph Neural Networks (GNNs) for predicting the electronic band gaps of Metal-Organic Frameworks (MOFs)—a class of highly tunable porous materials with applications in catalysis, gas storage, optoelectronics, and sensing. Traditional methods like Density Functional Theory (DFT) are computationally expensive and impractical for high-throughput screening. Machine learning offers a promising alternative, but the choice between feature-based models and structure-aware representation learning remains unresolved.

A rigorous comparative study is conducted between two state-of-the-art approaches: a highly optimized feature-based model trained on a comprehensive set of hand-engineered descriptors, and a Crystal Graph Convolutional Neural Network (CGCNN) that learns directly from atomic structures.

Key contributions of this work include the curation of a novel, feature-rich dataset integrating global geometric and atomistic electronic properties; the establishment of a strong, interpretable baseline model; and the implementation of an advanced GNN architecture. The comparative analysis demonstrates the relative advantages of each paradigm, with the GNN approach showing a marked improvement in predictive performance. Furthermore, model interpretation techniques were employed to extract physically meaningful insights into the structural and chemical factors governing band gaps in MOFs.

This work underscores the significant potential of GNNs in capturing complex structure-property relationships and contributes to the development of more accurate and efficient computational tools for materials discovery.

Keywords: Graph Neural Networks, Metal-Organic Frameworks, Band Gap Prediction, Machine Learning, Materials Informatics, XGBoost, CGCNN.

Keywords: Graph Neural Networks, Material Characterization, Machine Learning, Deep Learning

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Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
GNN	Graph Neural Network
ML	Machine Learning
DL	Deep Learning
MOF	Metal-Organic Framework

Chapter 1. Introduction

1.1 The Promise and Challenge of Metal-Organic Frameworks

[3],[4],[5] One of the most fascinating and adaptable material classes to be discovered in recent decades is Metal–Organic Frameworks (MOFs). Metal ions or clusters (which act as nodes) and organic ligands (which act as linkers) self-assemble to form these crystalline compounds, which result in highly porous, adjustable, and modular structures [3]. MOFs are ideal candidates for a wide range of applications, including gas storage and separation (e.g., carbon capture, hydrogen storage) [4], catalysis [5], chemical sensing [6], drug delivery [7], and especially in optoelectronics and energy applications like photovoltaics, light-emitting diodes (LEDs), and photocatalysis [8].

MOFs' near-infinite tunability and precise architecture are what really make them brilliant. Chemists can carefully engineer the material's properties in a way that is unmatched by most other material classes by merely altering the metal cluster (such as Zn, Cu, Zr, or Cr) or the organic linker (such as dicarboxylates, triazolates, or imidazolates). This process of putting together molecular building blocks into predefined ordered structures is frequently referred to as "reticular chemistry." Because of this, frameworks with particular pore sizes, surface areas, and chemical functionalities can be systematically designed, transforming them from a single material into a vast and designable platform for cutting-edge applications.

One essential characteristic that determines a material's electronic and optical behavior is its electronic bandgap. It is the difference in energy between the conduction band, which is empty, and the valence band, which is filled with electrons. This characteristic establishes whether a substance is an insulator (large gap), semiconductor (small to moderate gap), or metal (no gap). MOFs' bandgap needs to be carefully designed in order for them to be used in electronic devices. A bandgap that can be excited by visible light is necessary for photocatalysis, for example, whereas some sensing applications might need a particular gap to enable charge transfer upon analyte binding. [6].

There is more to the electronic band gap than just a straightforward metal-insulator binary classification. Critical performance metrics in devices are determined by its magnitude and nature (direct vs. indirect). For example, a band gap of about 1.5 eV is optimal for photovoltaics in order to capture a wide range of sunlight. The energy positions of the valence and conduction bands must straddle the redox potentials of water, and the band gap must be sufficiently narrow to be excited by visible light (¿1.23 eV, but usually ¡3.0 eV) for photocatalysis, such as water splitting. A slight alteration in the band gap upon analyte binding can serve as a highly sensitive transduction mechanism in sensing.Therefore, the accurate prediction and precise engineering of the band gap are not merely academic exercises but are fundamental to unlocking the functional potential of MOFs in next-generation technologies.

1.2 The Computational Bottleneck in MOF Discovery

The conventional method of finding new MOFs with desired characteristics mainly depends on labor-intensive, costly, and slow experimental synthesis and characterization. Predicting material properties *in silico*, directing experimental efforts, and speeding up the discovery cycle are all made possible by computational techniques, especially those based on quantum mechanics like Density Functional Theory (DFT).

Despite its strength, DFT's computational cost is too high to screen the enormous, nearly limitless chemical space of potential MOFs. For instance, a moderately complex MOF may require thousands of CPU hours to perform a single HSE06 hybrid functional calculation, which yields high-fidelity bandgap values that are more in line with experimental findings. It is computationally impossible to screen thousands of possible candidates using this level of theory. Citation: Jain (2013). This limits our ability to fully explore the potential of MOFs by causing a significant bottleneck in the high-throughput discovery pipeline.[9, 10].

1.3 The High-Throughput Screening Paradigm

Traditional serendipitous discovery is pointless when considering the vastness of MOF chemical space, which is thought to contain millions or even billions of plausible structures. The high-throughput computational screening (HTCS) paradigm was born out of this difficulty. In the HTCS workflow, a large database of candidate structures is created or gathered, their properties are computationally simulated using techniques such as DFT, and they are subsequently filtered and ranked according to desired performance metrics. By directing experimental efforts towards the most productive regions of chemical space, this method has demonstrated remarkable success in identifying promising candidates for gas storage, separation, and other applications.

Nonetheless, quantum mechanical simulations serve as the foundation for this paradigm, which is crucial and computationally costly. Depending on the size of the system and the level of theory applied, a DFT calculation can take hours or weeks to complete for each data point displayed in a high-throughput screen. This leads to a serious bottleneck. We are completely incapable of calculating the properties of millions of MOFs, but we can do so for thousands. Because of this restriction, the great majority of the hypothetical MOF space is still entirely unexplored and unmapped. Despite its strength, the HTCS approach only scratches the surface of what can be done; if the computational bottleneck can be removed, there is a huge potential for accelerated discovery.

1.4 Machine Learning in Materials Science: A Primer

By using available data to learn the intricate relationship between a material's structure and properties, machine learning (ML) provides a potent remedy for the computational scaling issue. An ML model learns the underlying patterns from a training set of previously calculated materials rather than solving the intricate Schrödinger equation for each new candidate. When compared to DFT, the model's computational cost is almost zero, and once trained, it can predict properties for novel, unseen structures in milliseconds.

The ability to translate a complex, frequently non-periodic atomic structure into a numerical format that an algorithm can understand is the foundation of any machine learning application in materials science. Two prevailing philosophies are the result of this challenge. The first is feature-based or descriptor-based learning, which uses human creativity to identify a collection of numerical values (features) that best represent the salient features of the content. The quality and completeness of these human-engineered descriptors then have a significant impact on the model's performance. In the second paradigm, known as representation learning, the model is given a more basic representation of the structure, such as a list of atomic coordinates and elements, and is entrusted with learning both the structure-property relationship and the best way to represent the input data itself. Graph Neural Networks are the pinnacle of this approach for materials, as they can operate directly on the native graph structure of a crystal.

1.5 Machine Learning as a Paradigm Shift

[1, 11] A paradigm shift toward data-driven approaches has been sparked by the shortcomings of conventional computational methods. A potent substitute is provided by machine learning (ML), which uses available data to understand the intricate, non-linear relationships between a material's structure, composition, and properties. An ML model can predict in milliseconds after it has been trained, negating the need for costly DFT calculations for novel candidate materials.[1, 11].

With ML models effectively forecasting a broad range of properties—from formation energies and bandgaps to gas adsorption capacities and mechanical properties—the field of materials informatics has experienced tremendous growth. These models fall into two general philosophical categories. [1, 11, 12]:

1. **Feature-Based Models:** These rely on features (descriptors) that are human-engineered and represent the material. The caliber and thoroughness of these features have a significant impact on the model's performance.

2. **M**odels of Representation Learning: These models, especially Graph Neural Networks (GNNs), can potentially capture subtle patterns that are hard to encode manually because they learn their own representations directly from the material's raw, atomic-level structure [13–15].

1.6 Knowledge Gap and Research Questions

Although both strategies have shown promise, a crucial query still needs to be answered: Does the additional complexity of representation learning with GNNs offer a substantial and meaningful advantage over a well-built, feature-based model for predicting electronic properties like bandgaps in complex, porous materials like MOFs? [16]

Even though a lot of research has shown that GNNs can perform better than basic baseline models (like linear regression or small neural networks) on a small set of features, these comparisons are frequently unfair and inconclusive. Can a GNN beat a highly optimized, cutting-edge classical model trained on an extensive collection of carefully constructed features? This is a more exacting question. It is challenging to determine whether the improvement is due to the GNN's superior architecture or just a more advanced overall modeling technique because many published works lack this strong baseline. In order to ensure that any outperformance by the GNNs is a genuine testament to the worth of its learned structural representations, this thesis builds what is arguably the strongest feature-based model for this task.

Many studies introduce new GNN architectures and demonstrate their superiority over simple baselines. However, a rigorous and fair comparison against a *strong*, highly optimized feature-based baseline is often lacking. This thesis seeks to fill this gap by conducting a controlled, in-depth comparative study [13–15].

Numerous studies present novel GNN architectures and show how they outperform basic baselines. However, there is frequently a lack of a thorough and equitable comparison against a *strong*, highly optimized feature-based baseline. By performing a thorough, controlled comparative analysis, this thesis aims to close this gap.

The following are the main research questions that are addressed:

1. RQ1: Using a state-of-the-art, feature-based model (such as XGBoost) with an extensive, hybrid feature set derived from domain knowledge, what is the predictive performance ceiling for the bandgap property?
2. RQ2: Does a graph-based representation learning model (e.g., CGCNN) that learns directly from atomic structure improve predictive performance over the strong feature-based baseline in a way that is both statistically significant and chemically meaningful?
3. RQ3: In terms of accuracy, interpretability, computational cost, and data requirements, what are the relative benefits and drawbacks of each paradigm? [17]

In order to answer RQ3, a comprehensive comparison that goes beyond basic accuracy metrics is required. MAE and R2 are important, but they don't provide the whole picture. Scientific discovery depends on a model's interpretability; a feature-based model, such as XGBoost, offers distinct feature importance scores, whereas a GNN necessitates more sophisticated methods to comprehend its predictions. Another important consideration is computational cost: a GNN's prediction time is comparable to XGBoost's, but its training time can be orders of magnitude longer. Additionally, different data sets are needed for training; GNNs typically need more data, but they may be able to generalize better to areas of chemical space that are not visible. A materials scientist must be aware of these trade-offs in order to select the best instrument for a particular issue.

1.7 Scope and Delimitations

The prediction of the electronic band gap is a primary target property in this thesis. The band gap is a fundamental electronic property that is both computationally costly to calculate with high accuracy and crucial for the targeted applications in energy and optoelectronics, even though MOFs have many other desirable properties (such as gas uptake, mechanical stability, and catalytic activity). Only structures from the QMOF database, which offers a reliable and superior collection of DFT-relaxed structures and computed properties, are included in the study. While this guarantees consistency in the data, it might restrict the models' direct application to experimentally synthesized MOFs, which might have solvents, defects, or other flaws not found in the computational idealization.

Additionally, this study compares two particular implementations of the feature-based and representation learning paradigms, XGBoost and CGCNN. Although there are numerous other model types and GNN architectures (such as MEGNet, ALIGNN, and SchNet), the goal of this study is to present a thorough and rigorous comparison of these two popular methods rather than a general but superficial overview of the whole machine learning field. However, it is anticipated that the knowledge acquired will be transferable to a more comprehensive comparison between structure-aware GNNs and meticulously designed feature models.

1.8 Thesis Objectives and Contributions

The main goal of this thesis is to create and implement a strict methodological framework for bandgap prediction in MOFs that is intended to impartially assess the two rival machine learning paradigms. [1, 11]. The key contributions are:

- **Comprehensive Data Curation:** From the QMOF database, a new feature-rich dataset (`qmof_with_dft_aggregates`) was created by combining crystallographic symmetry, global geometric descriptors, and—most importantly—statistical aggregates of atom-site DFT properties (charges, spin densities) [9, 16, 18].

- **Strong Feature-Based Baseline:** Advanced GNN Implementation: An effective graph-caching system and global feature integration have been added to the Crystal Graph Convolutional Neural Network (CGCNN) implementation. [19, 20].
- **Advanced GNN Implementation:** An effective graph-caching system and global feature integration have been added to the Crystal Graph Convolutional Neural Network (CGCNN) implementation. [13–15].
- **Rigorous Comparative Analysis:** a thorough assessment that takes into account interpretability, robustness, and computational efficiency in addition to accuracy metrics, offering a comprehensive perspective of the trade-offs involved. [17].
- **Physical Insight:** Utilizing model interpretation tools (feature importance, SHAP, and gradient attribution) to derive scientifically significant information about the chemical and structural elements controlling bandgaps in MOFs.[17].

1.9 Thesis Outline

This thesis is structured to guide the reader through the entire research process:

- **Chapter 2: Related Work** Discusses feature engineering, graph-based learning, and earlier research on MOF property prediction as it relates to the development of machine learning in materials science. [1, 11].This section will also chart the field's historical development from the emergence of materials informatics to the early chemistry descriptor-based QSPR models. The various descriptor families—compositional, geometric, and electronic—as well as the cheminformatics idea of molecular fingerprints will be reviewed. The revolution brought about by deep learning and graph neural networks will then be discussed, along with important architectures like CGCNN, MEGNet, and ALIGNN and how they are specifically used to predict MOF properties.
- **Chapter 3: The QMOF Dataset** Explains the origin of our data, its composition, and the thorough curation process used to produce a dataset that is ready for machine learning. [9, 16, 18] along with offering a thorough examination of the data source. The relaxation process, the various theoretical levels (PBE vs. HSE06), and the high-throughput DFT calculation process will all be covered in detail. The database's structure, the different properties that are available, and the preliminary data cleaning procedures needed to get it ready for machine learning will all be covered in this chapter.
- **Chapter 4: Exploratory Data Analysis** Provides a comprehensive statistical analysis of the dataset, highlighting important trends, connections, and difficulties that influenced the preprocessing approach. Will also act as a vital link between modeling and raw data. It will provide a comprehensive statistical analysis that includes the distribution of the target variable (HSE06 band gap), missing data analysis, feature correlation, and outlier detection. The preprocessing and feature selection techniques used in later chapters will be directly influenced by the EDA.

- **Chapter 5: Feature-Based Baseline** Explains the preprocessing pipeline, the training/tuning of the strong baseline model, and the philosophy underlying the hybrid feature set. [19, 20].It is going to be dedicated to the philosophy of feature engineering. It will justify the selection of each descriptor family (geometric, electronic aggregates, symmetry, chemical fingerprints) based on physical principles. It will then detail the comprehensive preprocessing pipeline, including handling missing data, encoding categorical variables, and mitigating skewness and multicollinearity. The chapter will conclude with the methodology for hyperparameter tuning and the evaluation of the final XGBoost model.
- **Chapter 6: CGCNN Model** describes the GNN theory, the CGCNN architecture in detail, and the specifics of our improved model's implementation. [13–15].
- **Chapter 7: Results and Discussion** Evaluates both models' performance, looks for mistakes, and interprets the findings to make physical inferences. This section makes the switch to representational learning. It will describe the message-passing framework and the theoretical underpinnings of graph neural networks. A thorough explanation of the CGCNN architecture will be given, including how atom-level data is combined to create a global prediction, how convolutional layers work, and how crystal graphs are built from CIF files. The training protocol, improvements (such as the integration of global features), and specific implementation details will also be covered in this chapter.
- **Chapter 8: Conclusion and Future Work** highlights the main conclusions, talks about the study's shortcomings, and suggests exciting avenues for further investigation. This expands and compares the two models' performance side by side on a test set that has been held out. In addition to examining performance across various band gap regimes and examining particular instances of success and failure, the analysis will go beyond aggregate metrics. The interpretability analysis, which will be a major component of this chapter, will use techniques such as gradient attribution and SHAP values to extract physical insights and comprehend what each model has learned about structure-property relationships in MOFs.

  Conclusion and Prospects The work will summarize the main conclusions and provide a conclusive response to the original research questions. The limitations of the current work will be discussed, and specific research directions will be suggested, including the use of more sophisticated GNN architectures, the investigation of transfer learning from larger materials datasets, and the extension of the framework to predict additional properties or classes of porous materials.

Chapter 2. Related Work: State-of-the-Art in Cheminformatics and Data Science for Materials

2.1 The Confluence of Cheminformatics and Materials Informatics

From being primarily an empirical discipline to one that is increasingly driven by data-driven discovery, materials science has experienced a significant transformation. This change has been driven by the convergence of two strong fields: **materials informatics**, which uses data science to discover new materials, and **cheminformatics**, which uses computational methods to solve chemical problems. Despite cheminformatics's historical emphasis on small molecules and drug discovery, its concepts, instruments, and representations have shown remarkable efficacy in modeling complex crystalline materials, such as Metal-Organic Frameworks (MOFs) [1]. Tracing the development from basic descriptors to deep learning, this chapter offers an extensive overview of the state-of-the-art approaches at this intersection.

These fields' mutual challenge—how to numerically represent chemical systems for computational analysis—is what gave rise to their synergy, which is not just coincidental. In order to facilitate virtual screening and quantitative structure-activity relationship (QSAR) modeling in drug discovery, cheminformatics developed powerful fingerprinting algorithms and canonical representations such as the Simplified Molecular Input Line Entry System (SMILES). In contrast, materials science initially used more macroscopic descriptors because it dealt with extended crystalline solids. The critical mass of structured data made possible by the rise of massive computational databases such as the Cambridge Structural Database and the Materials Project enabled materials science to embrace and modify the advanced data-driven approaches that were first introduced in cheminformatics.This confluence has created a new interdisciplinary playbook where concepts like molecular fingerprints are applied to crystalline unit cells, and QSAR evolves into QSPR (Quantitative Structure-Property Relationship) for materials

2.2 Historical Context: The Descriptor Paradigm

The foundation of conventional machine learning in materials science is the descriptor paradigm. It works on the simple premise that intricate atomic systems can be reduced to a fixed-length vector of numerical features that encapsulate their composition and structure. The selection and creation of these descriptors are the only aspects of this paradigm that are both art and science. Physically meaningful, computationally cheap, invariant to irrelevant transformations (such as unit cell rotation), and, most importantly, correlated with the target property are all requirements for effective descriptors. As a result, descriptor families have proliferated, each intended to capture a distinct facet of a material's identity.

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Based on the idea that intricate atomic systems can be reduced to a fixed-length vector of numerical features that encapsulate their fundamental composition and structure, the descriptor paradigm serves as the cornerstone of conventional machine learning in materials science. The selection and design of these descriptors, which must be physically meaningful, computationally cheap, invariant to irrelevant transformations, and correlate with the target property, constitute the art and science of this paradigm. As a result, descriptor families have proliferated, each intended to capture a distinct facet of a material's identity.

2.2.1 Composition-Based Descriptors

The most basic of these are composition-based descriptors, which function on the premise that bulk properties are determined by the average behavior of constituent elements. These include estimates of mixing enthalpy, weighted averages of elemental properties, electronegativity differences to capture charge transfer, and ratios such as oxygen-to-metal, which go beyond basic statistics. Their incapacity to differentiate between distinct polymorphs, such as diamond and graphite, which have the same composition but radically different properties, is a fundamental flaw, despite their strength for properties like bulk modulus. Although they are an essential place to start, they fall short of fully encapsulating the intricacy of structure-sensitive characteristics like band gap.

Assuming that bulk properties are determined by the average behavior of constituent elements, these descriptors are the most basic. Weighted averages, which weight elemental properties by atomic fraction; electronegativity differences, which capture the tendency for charge transfer and ionic character; estimates of mixing enthalpy based on the thermodynamics of mixing; and straightforward but effective ratios, such as oxygen-to-metal for oxide stability, are examples of more complex composition-based features that go beyond basic statistics. Their primary shortcoming is the "diamond vs. graphite" problem, which is that they are unable to differentiate between various polymorphs or allotropes due to their identical composition, even though they are strong for specific properties like bulk modulus or average formation energy. They serve as a foundation, but they fall short of fully encapsulating the intricacy of structure-sensitive characteristics such as band gap.

2.2.2 Geometric and Topological Descriptors

The spatial arrangement of atoms in crystalline materials is captured by geometric descriptors. This comprises lattice parameters, density, and unit cell volume, among other global geometric features. Pore Limiting Diameter (PLD), Largest Cavity Diameter (LCD), void fraction, and accessible surface area are important porosity metrics for porous materials like MOFs and zeolites. The "stage" for chemistry is established by these descriptors. In order to characterize the network's connectivity, topological descriptors frequently reduce a MOF structure to its underlying net topology (such as pcu or dia) and use this as a categorical feature. Although effective for categorization, this abstraction may obscure crucial chemical information.

Geometric descriptors are crucial for porous materials such as MOFs. Zeo++ and PoreBlazer are two tools made especially to compute important metrics other than volume and density. These include the Pore Size Distribution (PSD), a histogram of pore volumes as a function of pore width that offers a more nuanced view than a single PLD or LCD value; the Accessible Surface Area (ASA), which is the surface area accessible to a probe molecule of a particular size (e.g., N for surface area, H for hydrogen storage); and channel dimensionality, which indicates whether the pores are 1D, 2D, or 3D and has a substantial impact on molecular diffusion. Topological descriptors abstract the MOF to its underlying net, a mathematical graph where metal clusters and organic linkers are simplified to nodes. While classifying a MOF as having a pcu (primitive cubic) or dia (diamond) topology is useful for categorization, this high-level abstraction discards all chemical information. A pcu net could be built from zinc-based clusters and carboxylate linkers or from copper and azoles, resulting in materials with wildly different chemical and electronic properties, a difference this descriptor cannot capture.

2.2.3 Electronic Structure Descriptors as Proxies

A new class of descriptors—statistical aggregates of atom-resolved electronic properties—arose with the growth of high-throughput computational chemistry. This work makes extensive use of this potent idea. Statistics such as mean, median, standard deviation, and quartiles are computed for each atom in the unit cell for a particular property (e.g., partial charge, magnetic moment). One effective indicator of the material's overall ionicity and electronic heterogeneity is the standard deviation of partial charges. This method successfully produces a "fingerprint" of the electron density distribution, offering a comprehensive feature set that transcends basic geometry and is informed by physics.

This class of descriptors is a major advancement. They employ less expensive, lower-level DFT computations (such as the PBE functional) to produce potent features for forecasting more costly properties (such as HSE06 band gaps) rather than disregarding or merely using DFT results as a target. The fundamental idea is that the key physics governing the target property is encoded in the distribution of electronic properties within the unit cell. The Standard Deviation of Partial Charges $pbe_ddec_charge_std, forinstance, isapotentdescriptor; ahighvaluesuggestsamaterialwithsubstantialele$

2.2 The Cheminformatics Contribution: Representing Chemistry

The weak encoding of particular chemistry, such as the type of functional groups, aromaticity, and metal-linker interactions, is a major drawback of purely geometric or electronic descriptors. Here is where cheminformatics tools have been extremely helpful. A common method is to use a fixed-length binary vector, or fingerprint, to represent a molecule. Each bit of this vector indicates whether a specific molecular substructure is present or absent. The Extended-Connectivity Fingerprint (ECFP), sometimes referred to as the Morgan fingerprint, is the most widely used algorithm. It is possible to encode the local chemical environment in a manner that is sensitive to functional groups and connectivity while remaining invariant to the precise atomic coordinates by applying ECFP to the SMILES strings of a MOF's organic linkers. This enables models to learn, for example, that an azole linker affects selectivity or that a carboxylate group tends to lower the bandgap. The paradigm is now changing from feature engineering to representation learning, where we let the model learn the best representation from raw or minimally processed data rather than pre-defining what is important.

Extended-Connectivity Fingerprints (ECFP) are a potent link between materials and molecular informatics when applied to MOF linkers. It is possible to encode the local chemical environment in a manner that is sensitive to the presence of functional groups and topological connectivity while remaining invariant to the precise atomic coordinates by creating a fingerprint from the SMILES string of an organic linker. This enables machine learning models to recognize that a carboxylate group, for example, is frequently linked to specific electronic characteristics. Nevertheless, this strategy has built-in drawbacks. A single linker molecule is used to create the fingerprint, with no regard for how well it coordinates with the metal node. It is unable to account for the geometric limitations imposed by the framework or the impact of the metal center on the electronic structure of the linker (for example, through back-bonding). Additionally, it disregards the relative spatial arrangement of various functional groups within the pore and treats the fingerprint as a "bag of words." Despite these drawbacks, their proven ability to predict outcomes indicates that, even with imperfect representation, local chemical identity is a significant factor.

2.3 The Deep Learning Revolution: Graph Neural Networks

A graph is a molecule's or crystal's natural data structure. The most important recent developments in the field are the result of this realization. Message passing is the foundation of how Graph Neural Networks (GNNs) work. By combining "messages" from nearby nodes, nodes (atoms) in each layer update their hidden states. The representation of a node after multiple layers contains details about its local chemical environment. Without direct human assistance, this procedure enables the model to learn hierarchical features, ranging from local bonding environments to longer-range interactions.

A groundbreaking design that tailored the GNN concept for periodic crystals was the Crystal Graph Convolutional Neural Network (CGCNN). [14]. With atoms acting as nodes and edges connecting them within a cutoff distance, it depicts a crystal as a graph. Both node (element) and edge (distance) features are incorporated into its message-passing function. On extensive benchmarks, CGCNN outperformed conventional models, demonstrating that GNNs could automatically discover chemically significant patterns without the need for human-defined features. Important innovations have been introduced by later architectures. In order to predict properties under various conditions, some models integrated global state attributes (such as temperature and pressure) into the readout function. Others realized that bond angles, in addition to bond lengths, determine many material properties. These models construct an additional graph to explicitly incorporate angular information, leading to state-of-the-art accuracy on a wide range of properties.

Two steps can be used to formalize message passing in a CGCNN, the fundamental function of a GNN, for every node at every layer. The first is the Message Function, which computes a message for every node based on each of its neighbors. The neighbor's current state and the characteristics of the edge that connects them determine this message. The node aggregates all incoming messages (for example, by using sum, mean, or max) and then uses the aggregated message and its previous state to update its own state. This is the second function. Information from each of a node's multi-hop neighbors is included in its representation after a number of layers.A groundbreaking architecture that specifically adapted the GNN concept for periodic crystals was the Crystal Graph Convolutional Neural Network (CGCNN). With architectures that included more physical inductive biases after CGCNN, the field quickly progressed. By using global state variables (such as temperature and pressure) as inputs, MEGNet (Materials Graph Network) enabled a single model to forecast properties under various circumstances. By creating a second graph in which nodes stand in for bonds from the original graph and edges for angles between bonds, ALIGNN (Atomistic Line Graph Neural Network) addressed CGCNN's drawback of ignoring angular information and explicitly included important three-body interactions.

2.4 Prominent Applications to Metal-Organic Frameworks

Thanks to the availability of extensive databases, the application of these cutting-edge methods to MOFs has flourished as a subfield. Training complex models has been made possible in large part by the development of consistently computed databases. The necessary information for feature-based and graph-based methods is provided by these repositories. Key application areas have been the focus of predictive studies. The most popular use has been gas adsorption and separation, where models forecast selectivity and uptake. Another crucial task is predicting formation energy to evaluate thermodynamic stability. The main focus of this thesis is bandgap prediction, which is a more recent and difficult focus. For example, the QMOF database has established itself as a common standard for these kinds of electronic property prediction tasks. [16]. The ultimate objective is to create novel MOF structures with desired properties through inverse design, which goes beyond prediction. Generative models, which learn a latent space of MOF structures, are commonly used to address this. Compared to predictive modeling, this field is still in its infancy, despite its promise.

The development of the field has been reflected in the application of ML to MOFs. Given that the "physical" structure of the MOF has a significant impact on these properties, it makes sense that early research concentrated on forecasting gas adsorption (such as CO uptake and CH working capacity) using geometric descriptors like surface area and pore volume. Models started to address increasingly complex chemical properties like catalytic activity and selectivity as feature sets expanded to include electronic descriptors and cheminformatics fingerprints. Because it depends so heavily on the exact electronic structure, band gap prediction is a modern frontier challenge that calls for models that can deeply integrate geometric, chemical, and electronic information. The shift from feature-based models to GNNs is particularly pronounced here, as GNNs natively blend all these information types directly from the atomic structure, making them ideally suited for this complex task.

2.5 The Critical Role of Interpretability and Uncertainty

Gaining scientific trust and insight requires an understanding of the predictions made by increasingly complex models. Techniques for Explainable AI (XAI) are being used more and more. Scores for intrinsic feature importance are provided by tree-based models. Model-agnostic techniques such as SHAP (SHapley Additive exPlanations) provide both local and global interpretability by allocating an importance value to each feature for a particular prediction. [17]. By linking model behavior to chemical intuition, gradient-based attribution techniques for GNNs can reveal which atoms or bonds had the greatest influence on a prediction. Moreover, an estimate of a prediction's confidence is necessary for it to be actionable. Prediction uncertainty is measured *using methods such as ensemble methods. This makes it possible for researchers to give high-confidence predictions' experimental validation top priority, which is essential for autonomous discovery loops.*

Accuracy alone is not enough in scientific machine learning; understanding and discovery are frequently the main objectives. A black box model that makes perfect predictions provides little scientific information. Researchers can use interpretability tools to find errors and flaws in the training data or descriptor set, generate new hypotheses by demonstrating that a particular, underappreciated sub-structure is a strong predictor, and validate the model by confirming that the identified important features match accepted physical principles. Moreover, an estimate of a prediction's confidence is necessary for it to be actionable.By differentiating between aleatoric uncertainty (irreducible "noise" present in the data) and epistemic uncertainty (uncertainty in the model resulting from a lack of training data), Uncertainty Quantification (UQ) distinguishes between a trustworthy predictive engine and a valuable research tool. A model can accurately detect when it is making a prediction outside of its knowledge domain by measuring epistemic uncertainty. This is essential for directing active learning loops, which maximize resource allocation efficiency in a high-throughput discovery pipeline by allowing the model to choose the next most instructive data points to be computed.

2.6 Positioning and Contribution of This Work

This rich and quickly changing landscape is where this thesis is situated. Our study compares two prevalent and representative paradigms in a thorough, controlled, and exacting manner: :

1. A ***state-of-the-art feature-based approach****: A cutting-edge feature-based method: This is the pinnacle of the descriptor paradigm, utilizing the entire suite of cheminformatics and materials informatics tools in a highly optimized XGBoost framework. This baseline is robust and significant.*

2. An ***implementation of a foundational GNN architecture (CGCNN)****: The representation learning paradigm is exemplified by this. We add useful enhancements, such as global feature integration, to the standard implementation.*

This rich and quickly changing landscape is where this thesis is situated. It contributes by offering a conclusive standard in the continuing discussion between representation-learning and feature-based methodologies. It is challenging to determine the actual value added by the architectural complexity because so many prior studies introduce new GNN architectures and show their superiority over basic or antiquated baselines. Our research tackles this by building a genuinely robust and representative baseline: a "best-in-class" descriptor-based model that makes use of an extensive hybrid feature set that combines local chemical, aggregated electronic, and global geometric descriptors within an extremely well-optimized XGBoost framework.

It is challenging to determine the actual value added by the architectural complexity of a new, complex GNN because many prior studies compare it to a basic or antiquated baseline. Building a genuinely robust and representative baseline is what we have to offer. This makes it possible to evaluate the question more conclusively: Does the additional complexity and computational expense of graph-based representation learning offer a substantial and meaningful advantage over what is currently possible with expert-designed features and potent classical ML for predicting electronic properties in MOFs? The response to this query gives the community important information about the real-world trade-offs between these two potent strategies.

This makes it possible to evaluate the central question more definitively: does the computational expense and additional complexity of graph-based representation learning offer a substantial and meaningful advantage over what is currently possible with expert-designed features and potent classical ML for predicting electronic properties in MOFs? The response to this query gives the community important information about the real-world trade-offs between these two potent strategies.

Chapter 3. Methodology and Implementation

3.1 Introduction to Methodology

3.1.1 Introduction

A major challenge at the nexus of machine learning and computational materials science is the precise prediction of electronic bandgaps in Metal-Organic Frameworks (MOFs). Photovoltaics, chemical sensing, light-emitting diodes (LEDs), and photocatalysis are just a few of the many applications that depend on a material's bandgap, a fundamental electronic property that determines whether it behaves as a conductor, semiconductor, or insulator [1] The intricate, non-linear interaction between atomic composition, crystalline structure, porosity, and electronic interactions makes it notoriously difficult to predict bandgaps with high accuracy, despite their critical importance..

Computational materials discovery has relied heavily on conventional, first-principles quantum mechanical simulations, especially those based on Density Functional Theory (DFT). High-fidelity results can be obtained using techniques like the Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional, but they are computationally prohibitive and frequently require thousands to millions of CPU hours for screening large material databases. [2]. They are not feasible for the high-throughput screening (HTS) required to traverse the nearly limitless chemical space of potential MOFs due to this computational bottleneck. A paradigm shift toward data-driven machine learning (ML) models has been sparked by this limitation. These models can learn the intricate mapping from material structure to properties from available data, providing precise predictions at a fraction of the computational cost. [11].

The thorough methodology used for this thesis, "Material Characterization using Graph Neural Networks," is described in this chapter. Predicting the HSE06 bandgap of a MOF given its structural and chemical information is the main challenge, which is expressed as a regression task. The methodology is intended to serve as a comparison of two potent but philosophically different approaches: an advanced Graph Neural Network (GNN) architecture called the Crystal Graph Convolutional Neural Network (CGCNN), which learns directly from the raw atomic structure, and a robust baseline model that uses the Extreme Gradient Boosting (XGBoost) algorithm on hand-crafted features. Beyond conventional feature-based machine learning, this dual-strategy approach is carefully crafted to rigorously quantify the value added by incorporating explicit structural information through contemporary representation learning techniques. The following sections describe the design's justification, the models' theoretical foundations, feature *engineering and representation learning concepts, and the stringent evaluation procedure used to guarantee results that are reproducible and safe for scientific use.*

3.1.2 Research Design and Comparative Approach

This thesis's research design is essentially comparative, contrasting a state-of-the-art deep learning architecture created especially for materials science with a state-of-the-art traditional machine learning model. In order to answer the main research questions, this design is essential:

1. *How accurately can bandgap predictions for MOFs be approximated using a highly optimized, feature-based model (XGBoost)?*
2. *Is there a statistically significant and chemically meaningful improvement in predictive performance when explicit, learned structural representations are incorporated using a graph-based model (CGCNN)?*
3. *n terms of accuracy, interpretability, computational cost, and data requirements, what are the relative benefits and drawbacks of each approach?*

3.1.2.1 The XGBoost Baseline: A Feature-Based Paradigm

To create a solid, comprehensible baseline, the XGBoost algorithm [15] was chosen. Building on the idea of gradient boosting, XGBoost is an ensemble technique that combines several weak prediction models (decision trees) to produce a strong learner. It is appropriate for this task because of a number of important features.:

- ***Handling of Heterogeneous Data**: It effectively handles a variety of data types, such as the continuous, discrete, and categorical characteristics frequently found in materials descriptors.*
- ***Integrated Regularization**: It effectively handles a variety of data types, such as the continuous, discrete, and categorical characteristics frequently found in materials descriptors. Given the high-dimensional feature space, a strong defense against overfitting is provided by the built-in L1 (Lasso) and L2 (Ridge) regularization as well as column and row subsampling.*
- ***Sparsity-Aware Learning**: It effectively handles a variety of data types, such as the continuous, discrete, and categorical characteristics frequently found in materials descriptors. It can handle sparse data matrices and missing values natively, which are frequently the result of one-hot encoding of categorical variables like space groups.*
- ***Interpretability**: The model provides intrinsic, impurity-based feature importances, allowing for a post-hoc analysis of which descriptors are most predictive of the bandgap.*

The performance of XGBoost is heavily dependent on the quality and comprehensiveness of the input features. This necessitates a sophisticated feature engineering pipeline, detailed in

Section 3.1.3, which integrates domain knowledge to create a hybrid set of descriptors encompassing global geometry, electronic structure aggregates, crystallographic symmetry, and local chemical information [20].

It effectively handles a variety of data types, such as the continuous, discrete, and categorical characteristics frequently found in materials descriptors. In materials informatics, this method exemplifies the conventional machine learning paradigm: expert knowledge is carefully incorporated into the model before learning through feature design. Finding intricate, non-linear relationships between these pre-established descriptors is the model's strong point. Its inability to access information not captured by these descriptors, like complex topological features of the MOF structure that are hard to quantify with simple scalars or long-range interactions, is its main flaw.

3.1.2.2 The CGCNN Model: A Representation Learning Paradigm

In contrast, the representation learning paradigm is embodied by the Crystal Graph Convolutional Neural Network (CGCNN) [14]. CGCNN learns a representation directly from the atoms and bonds that make up the crystal, rather than depending on pre-defined features.

A MOF's crystal structure is naturally represented as a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where:

- *$\mathcal{V}$ is the set of nodes (atoms).*
- *$\mathcal{E}$ is the set of edges (bonds or interatomic interactions within a specified cutoff radius).*

Each node i is attributed with a feature vector $\mathbf{z}_i$ encoding elemental properties (e.g., atomic number, group, period, electronegativity). Each edge between atoms i and j is attributed with the interatomic distance d_{ij}.

The CGCNN architecture operates through a series of message-passing or convolution steps:

1. ***Initialization**: A feature vector is initially assigned to each atom according to its elemental characteristics.*
2. ***Convolutional Layers**: The model collects data from nearby atoms in the crystal graph for every atom. "Messages" are passed along the edges to accomplish this. For atom i at layer t, a typical message-passing function is:*

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_{j \in \mathcal{N}(i)} \sigma\left(\mathbf{z}_{ij} \cdot \mathbf{W}_f^{(t)} + \mathbf{b}_f^{(t)}\right) \odot g\left(\mathbf{z}_{ij} \cdot \mathbf{W}_s^{(t)} + \mathbf{b}_s^{(t)}\right)$$

where $\mathbf{v}_i^{(t)}$ is the feature vector of atom i at layer t, $\mathcal{N}(i)$ is its neighborhood, $\mathbf{z}_{ij}$ is the edge feature, σ is a sigmoid function, g is a non-linear activation, $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters, and $\odot$ is the element-wise product.

3. ***Pooling**: The atom-level features are combined after multiple convolutional layers to create a fixed-length representation of the complete crystal structure, frequently using a normalized sum:*

$$\mathbf{v}_c = \frac{1}{N}\sum_{i=1}^{N}\mathbf{v}_i^{(T)}$$

where N is the number of atoms in the unit cell and T is the final layer.

4. ***Regression**: The final bandgap prediction is generated by passing this crystal-level representation through fully connected layers.*

Without being specifically instructed to search for them, this architecture enables CGCNN to automatically learn chemically relevant patterns, such as the impact of particular metal clusters or organic linker configurations on the electronic structure. Compared to feature-based models, it more accurately and impartially depicts the structure-property relationship.

3.1.2.3 The Rationale for Comparison

This thesis performs a controlled experiment by putting both models into practice. The XGBoost model is the best example of what can be accomplished with strong classical ML and expertly designed features. The potential of automated representation learning from unprocessed structural data is exemplified by the CGCNN model. The complexity of deep learning for this task is justified if CGCNN performs noticeably better than the robust XGBoost baseline. This shows that the model is effectively utilizing structural information that the feature-based approach cannot access. If they perform similarly, it indicates that the manually created feature set is already very informative, and other considerations like interpretability or processing power may influence the model selection. Making a significant contribution to the field requires this comparative design in order to go beyond merely implementing a new model and show its demonstrable superiority over accepted best practices.

3.1.3 Feature Engineering for the Baseline Model

Building a strong feature set is essential to the XGBoost baseline's performance. The hybrid feature engineering philosophy is used, which combines several descriptor families to produce a thorough representation of every MOF. This method is based on the knowledge that various scales of factors, such as local chemical bonding and global porosity, affect the bandgap.

3.1.3.1 Descriptor Families

The feature vector for each MOF is composed of four primary families of descriptors:

Global Geometric and Porosity Descriptors*The overall physical structure of the MOF, which affects characteristics like mass transport and quantum confinement, is captured by these descriptors. Important characteristics include:*

- ***Density*** *(`info.density`): The crystal density, which is frequently associated with mechanical stability and packing efficiency.*
- ***Pore Limiting Diameter (PLD)*** *(`info.pld`): One important metric for applications such as gas storage and separation is the diameter of the largest sphere that can diffuse through the framework.*
- ***Largest Cavity Diameter (LCD)****: The diameter of the largest sphere that can be enclosed within the framework.*
- ***Unit Cell Volume*** *(`info.volume`): The volume of the crystallographic unit cell.*
- ***Number of Atoms*** *(`num_atoms`): The number of atoms in the (conventional) unit cell, a crude measure of structural complexity.*

These characteristics influence elements like band dispersion and overlap, establishing the "stage" for electronic interactions.

Electronic Structure Aggregates*These features, which are proxies for the electronic environment inside the MOF, are derived from less expensive DFT calculations (typically PBE-level). They provide an overview of how important electronic characteristics are distributed among all of the atoms in the structure. The following statistical aggregates are computed for different charge and spin density populations (DDEC, Bader, CM5, etc.):*

- ***Mean*** *(e.g., `pbe_ddec_charge_mean`): The average value, indicating overall ionic character.*
- ***Standard Deviation*** *(e.g., `pbe_ddec_charge_std`): The spread of values, a proxy for electronic polarization.*
- ***Median****: The central value, robust to outliers.*
- ***25th and 75th Percentiles*** *(q25, q75): Describing the distribution tails.*

Aggregates for magnetic moments (`pbe_magmom`) are also included. These characteristics have a direct bearing on ideas that are essential to valence and conduction band alignment, such as electronegativity differences, ionicity, and electronic polarization.

Crystallographic Symmetry*Extended-Connectivity Fingerprints (ECFP4, also called Morgan fingerprints) are calculated to encode chemical information at the level of local atomic environments and functional groups [21]. These are circular fingerprints with a radius of two bonds that are hashed into a 512-bit vector and are created from the SMILES strings of the MOF's building blocks. Every bit denotes whether a particular molecular substructure is present or absent. This enables the model to capture chemistry that is lost in statistical aggregates and determine the effect of particular chemical motifs (such as a carboxylate group, an aromatic azole linker, or a particular metal cluster) on the bandgap.*

Local Chemical Fingerprints*Extended-Connectivity Fingerprints (ECFP4, also called Morgan fingerprints) are calculated to encode chemical information at the level of local atomic* *environments and functional groups [21]. These are circular fingerprints with a radius of two bonds that are hashed into a 512-bit vector and are created from the SMILES strings of the MOF's building blocks. Every bit denotes whether a particular molecular substructure is present or absent. This enables the model to capture chemistry that is lost in statistical aggregates and determine the effect of particular chemical motifs (such as a carboxylate group, an aromatic azole linker, or a particular metal cluster) on the bandgap.*

3.1.3.2 Preprocessing and Risk Mitigation

A rigorous preprocessing pipeline is implemented, treating data cleaning as a form of structured risk control:

- ***Label Integrity**: MOFs with missing or zero bandgap values are removed. **Ultra-Sparse Features**: Features with a high percentage (¿1%) of missing values are pruned.*
- ***Distributional Pathology**: Heavily skewed numerical features are transformed using* $\log(1+x)$ *or Yeo-Johnson power transforms [37]. Extreme outliers are winsorized (capped at the 1st and 99th percentiles).*
- ***High-Cardinality Categoricals**: Symmetry groups are one-hot encoded, keeping only the top-10 most frequent categories to avoid excessive sparsity.*
- ***Imputation**: Missing values in the remaining numerical features are filled using the median value.*
- ***Interaction Terms**: Simple multiplicative interaction terms between highly correlated features (e.g., `density * volume`) are engineered to help the model capture non-linear relationships.*

This process results in a robust, clean, and informative design matrix that allows the XGBoost model to perform at its best.

3.1.4 Representation Learning with Crystal Graphs

By learning directly from the atomic structure, the CGCNN model avoids the need for the extensive feature engineering mentioned above. This section describes the model mechanics and graph construction.

3.1.4.1 Crystal Graph Construction

For a given MOF's Crystallographic Information File (CIF), the crystal graph is built as follows:

1. ***Node Features** ($\mathbf{v}_i^{(0)}$): Based on its elemental characteristics, each atom i is given an initial feature vector. Usually derived from a pre-defined elemental embedding table or straight from periodic table data, these characteristics include atomic number, group, period, electronegativity, covalent radius, etc.*

2. ***Edge Definition***: *If the interatomic distance* d_{ij} *between two atoms* i *and* j *is less than a predetermined cutoff radius (e.g.,* 8Å*), then an edge is formed between them. The periodic nature of crystals is captured by this cutoff, which is large enough to include atom-to-atom interactions in nearby unit cells.*
3. ***Edge Features*** ($\mathbf{u}^{(0)}_{(i,j)_k}$)*: The feature for the edge between atoms* i *and* j *is typically a function of the interatomic distance. A common approach is to expand the distance into a set of Gaussian basis functions to represent a continuous range of bond lengths in a finite-dimensional vector:*

$$e_k = \exp\left(-\frac{(d_{ij} - \mu_k)^2}{\sigma^2}\right)$$

where μ_k *and* σ *are the center and width of the Gaussian basis functions.*

Without any human bias, this graph construction automatically includes all structural information, including any long-range interactions within the cutoff radius.

3.1.4.2 The CGCNN Architecture

Convolutional layers, which carry out message-passing functions, are the central component of the CGCNN [14]. For every atom i*, the operation at each layer* t *can be divided into two steps:*

1. ***Message Generation***: *For each neighbor* j *of atom* i*, a message vector* $\mathbf{m}^{(t)}_{ij}$ *is generated. This message is a function of the central atom's feature, the neighbor's feature, and the edge feature:*

$$\mathbf{m}^{(t)}_{ij} = MLP\left(\mathbf{v}^{(t)}_i \oplus \mathbf{v}^{(t)}_j \oplus \mathbf{u}_{(i,j)}\right)$$

where $\oplus$ *denotes vector concatenation and MLP is a multi-layer perceptron (a small neural network).*

2. ***Message Aggregation***: *The messages from all neighbors are aggregated (e.g., summed) and used to update the feature vector of atom* i*:*

$$\mathbf{v}^{(t+1)}_i = \mathbf{v}^{(t)}_i + \sum_{j \in \mathcal{N}(i)} \mathbf{m}^{(t)}_{ij}$$

Information from the local atomic environment is progressively incorporated into each atom's feature vector through this process. Following a number of these layers, the medium-range order in the crystal is effectively captured by the features of each atom, which contain information from atoms that are several bonds away.

3.1.4.3 Pooling and Readout

A rich hierarchy of structural and chemical information can be found in the atom-level feature vectors $\mathbf{v}^{(T)}_i$ *following* T *convolutional layers. These are combined using a permutation-invariant pooling operation, like mean pooling or weighted sum pooling, to create a single, fixed-length* *representation for the entire crystal:*

$$\mathbf{v}_c = \frac{1}{N}\sum_{i=1}^{N}\mathbf{v}_i^{(T)}$$

The structural features that are most important for predicting the target property are captured by this learned descriptor, the crystal-level embedding $\mathbf{v}_c$. The final bandgap prediction is then generated by passing it through a sequence of fully connected neural network layers:

$$\hat{E}_g = MLP_{regress}(\mathbf{v}_c)$$

The main benefit is that instead of depending on a predetermined, static set of features, the model learn this descriptor $\mathbf{v}_c$ optimized for the particular prediction task.

3.1.5 Advances in Material Representation Learning

Beyond the fundamental CGCNN architecture, the field of graph-based learning for materials is developing quickly. This thesis places its application of CGCNN in this larger framework, recognizing its advantages and disadvantages while suggesting potential paths forward.

3.1.5.1 Limitations of CGCNN and Subsequent Innovations

While powerful, CGCNN has known limitations:

- ***Isotropic Interactions**: Interatomic distances serve as the primary basis for message-passing, which treats all directions equally. Directional bonds and angular information, which are essential for characterizing coordination environments, are not specifically taken into consideration.*
- ***Lack of Global State**: Global factors that can impact material properties, such as temperature and pressure, were not included in the original model.*

Recent architectures have been developed to address these shortcomings:

- ***ALIGNN (Atomistic Line Graph Neural Network)** [22]: By creating a second graph in which nodes stand in for bonds in the original crystal graph and edges for angles between bonds, this model explicitly incorporates angular information. This enhances the accuracy of properties like formation energy and bandgap by enabling the model to learn from precise geometric features.*
- ***MEGNet (Materials Graph Network)** [15]: As an extra input to the graph readout function, this framework adds global state variables (such as temperature and pressure). This greatly increases the flexibility and physical grounding of a single model by enabling it to predict properties under various external conditions.*

Due to its foundational status, demonstrated performance on bandgap prediction tasks, and computational efficiency in comparison to more complex architectures like ALIGNN, CGCNN was selected as the representative GNN model for this thesis. It offers a precise and established standard for the representation learning methodology.

3.1.5.2 The Role of Transfer Learning

The ability of GNNs to facilitate transfer learning is a key benefit in materials science. General-purpose atomic and structural representations are learned by a model that has been pre-trained on a sizable, varied dataset of materials and their characteristics (such as the Materials Project database [2]). A smaller, task-specific dataset (such as a carefully selected collection of MOFs with HSE06 bandgaps) can then be used to refine this previously trained model. This paradigm is extremely powerful because it mimics how human scientists build upon prior knowledge by enabling the model to use knowledge from hundreds of thousands of structures to improve its performance on a target domain with limited data.

3.1.6 Evaluation Metrics and Model Robustness

Validating the predictive models requires a thorough and scientifically sound evaluation procedure. This evaluates generalization, dependability, and trustworthiness in addition to basic accuracy metrics.

3.1.6.1 Primary Performance Metrics

The following standard regression metrics are used to evaluate and compare the models:

- ***Mean Absolute Error (MAE):*** $\frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$*. This gives the average error in electronvolts (eV), providing an intuitive, scale-dependent measure of accuracy.*
- ***Root Mean Squared Error (RMSE):*** $\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$*. This penalizes larger errors more heavily than MAE, making it sensitive to occasional but significant prediction failures.*
- ***Coefficient of Determination*** (R^2)*: This measures the proportion of variance in the target variable that is predictable from the features. An* R^2 *of 1 indicates perfect prediction, while 0 indicates performance no better than predicting the mean.*

A held-out test set that is never used for model training or hyperparameter tuning is used to report these metrics.

3.1.6.2 Assessing Generalization and Robustness

Additional checks are carried out to make sure the models are learning strong relationships rather than merely memorizing the data:

- ***Train-Test Split****: To guarantee that the distribution of bandgap values in both sets is comparable, a stratified 80/20 split is employed.*

- ***Cross-Validation**: To avoid overfitting to a specific validation split, k-fold cross-validation (usually k=5) is used on the training set to perform hyperparameter tuning for the XG-Boost model.*
- ***Error Analysis**: To find systematic errors, predictions are examined. Does the model consistently perform worse for certain MOF types (such as those containing particular metals or linker types)? Essential diagnostic tools include residual plots and scatter plots of predicted versus true values.*

3.1.6.3 The Importance of Uncertainty Quantification

*For a prediction to be scientifically useful, an estimate of its confidence is often as important as the prediction itself. While not a primary focus of this initial study, the importance of Uncertainty Quantification (UQ) is acknowledged. Techniques like **Deep Ensembles** (training multiple models with different initializations) or **Monte Carlo Dropout** can be used with neural networks like CGCNN to provide a distribution of predictions rather than a single point estimate. For XGBoost, the inherent variability of the individual trees in the ensemble can be used to estimate uncertainty. Reliable UQ allows researchers to prioritize experimental validation for predictions made with high confidence, a crucial step towards autonomous materials discovery [36].*

3.1.7 Ethical and Sustainability Considerations

Beyond predictive accuracy, there are ramifications for the scientific use of machine learning. The following sustainable and ethical practices are taken into account when conducting this study:

3.1.7.1 Environmental Impact

It can be computationally and energy-intensive to train deep learning models, particularly large GNNs [25]. It's important to keep this expense in perspective, though, as the energy needed to train a model is frequently orders of magnitude less than the energy needed to perform high-throughput DFT calculations for an equivalent number of materials. Additionally, the model can make predictions almost instantly after training, essentially replacing millions of future CPU hours. Therefore, lowering the carbon footprint of computational research is a net benefit of an ML model's entire lifecycle in materials science [23].

3.1.7.2 Open Science and Reproducibility

This thesis adheres to the principles of Open Science to ensure reproducibility and foster collaboration:

- ***FAIR Data Principles**: The dataset used, derived from the QMOF database, is **F**indable, **A**ccessible, **I**nteroperable, and **R**eusable [24].*

- ***Code Availability**: All code for data preprocessing, model implementation (XGBoost and CGCNN), hyperparameter tuning, and evaluation will be made publicly available on a code repository like GitHub.*
- ***Transparent Reporting**: Full details of model architectures, hyperparameters, and evaluation metrics are provided in this thesis and any subsequent publications.*

This commitment ensures that the results can be verified and built upon by the wider scientific community.

3.1.7.3 Societal Benefit

Accelerating the discovery of MOFs for use in environmental remediation (such as carbon capture and water purification) and renewable energy (for example, photovoltaics and photocatalysis for fuel production) is the ultimate goal of this research. The goal of this work is to positively address urgent global challenges by offering effective computational tools for identifying promising materials.

3.1.8 Conclusion of Methodological Framework

A strong, comparative methodological framework for forecasting the electronic bandgaps of Metal-Organic Frameworks has been presented in this chapter. The dual-strategy approach is intended to provide a thorough evaluation of the benefits of structural representation learning by utilizing both a CGCNN deep learning model and a robust XGBoost baseline. Advanced feature engineering, sophisticated graph-based learning, a thorough evaluation protocol, and a dedication to moral and sustainable research practices are all included in the methodology. The application of this framework, the comparative study's findings, and a thorough analysis of the conclusions and their implications for the field of computational materials science will all be covered in the upcoming chapters.

3.2 Dataset

3.2.1 Introduction

A solid, carefully selected dataset is the cornerstone of any computational materials science investigation. The Quantum MOF (QMOF) database, a thorough compilation of properties for metal-organic frameworks that are derived from quantum mechanics, serves as the main source of data for this study. The QMOF dataset is explained in detail in this chapter, covering its history, the reasoning behind its creation, its complex file structure, and the wealth of data it contains across its different file formats. From the basic Crystallographic Information File (.cif) structures to the aggregated JSON metadata, we will examine the importance of each of its constituent parts and place this resource in the larger context of materials informatics, emphasizing its use in current literature and in solving important research issues.

A major step forward in the availability of large-scale, consistently computed quantum mechanical data for porous materials is the QMOF dataset, created by Rosen et al. [18]. Because of their enormous chemical and structural tunability, metal-organic frameworks pose a significant obstacle to conventional, experiment-led discovery. The QMOF database directly tackles this issue by offering a foundation for developing generalized machine learning models that can handle the intricate structure-property relationships present in this class of materials, in addition to predicting individual properties.

3.2.2 Dataset Provenance and Curation Rationale

Andrew S. Rosen and associates spearheaded a massive computational effort that resulted in the QMOF database. It was created primarily to fill a critical gap in the materials science data ecosystem: the absence of a sizable, internally consistent, and high-quality dataset of quantum mechanical properties for metal-organic frameworks. Although there are experimental databases of MOFs, they frequently have missing data, inconsistent measurement methods, and no underlying electronic structure information. Previous computational datasets frequently focused on a small number of properties and were either small or narrow in scope.

The enormous task of applying high-throughput density functional theory (DFT) calculations to a large collection of theoretical and experimentally realized MOF structures was undertaken by Rosen et al. [16]. Their process was made to be accurate and consistent. After being carefully deduplicated from pre-existing databases, the structures underwent a rigorous geometry optimization (relaxation) process using the Vienna Ab initio Simulation Package (VASP). This relaxation step is essential because it guarantees that the atomic configurations are at a local energy minimum, which leads to calculations of electronic properties that are more accurate and physically realistic. A wide range of properties, such as electronic band gaps, partial atomic charges, magnetic moments, and energetic stability, were calculated through a series of single-point calculations after relaxation.

As a result, the final dataset is a coherent and standardized resource rather than a mere collection of different computations. The QMOF database is particularly well-suited for machine learning, where consistency in the feature space is crucial for model performance and interpretability, because each entry benefits from the same computational parameters and theoretical level.

3.2.3 Dataset Architecture and File Structure

The QMOF dataset exhibits a hierarchical structure that is arranged to support both programmatic access and human inspection after acquisition and decompression. This architecture is clearly visible in the directory tree that is provided. (Figure 3.1).

```
relaxed_structure/
qmof_structure_data.json
scripts/
unrelaxed_structure.szip
qmof.csv
qmof.json
```

FIGURE 3.1: Top-level directory structure of the decompressed QMOF dataset.

The organization reflects a logical separation of concerns:

- *`relaxed_structure/`: This directory is the core of the dataset, containing the optimized crystal structures for each MOF.*
- *`unrelaxed_structure.szip`: This compressed archive contains the initial, pre-relaxation structures for reference and comparison.*
- *`qmof.json` and `qmof.csv`: These files contain the aggregated, tabular data—the calculated properties for each MOF—in JSON and CSV formats, respectively. They serve as the primary entry point for data analysis.*
- *`qmof_structure_data.json`: This file likely contains supplemental structural data or serves as an index linking MOF identifiers to their corresponding structure files.*
- *`scripts/`: This directory presumably contains utility scripts provided by the dataset authors for data loading, processing, or analysis.*

This structure follows a common pattern in computational materials science that strikes a balance between depth of information and usability for machine learning tasks by gracefully separating the computed properties (the '.json'/'.csv' files) from the raw structural data (the '.cif' files).

3.2.3.1 The Relaxed Structure Directory: A Repository of Atomic Configurations

The `relaxed_structure/` directory is arguably the most critical component, housing thousands of Crystallographic Information Files ('.cif'). Each '.cif' file corresponds to a single, unique MOF that has undergone DFT-based geometry optimization.

The International Union of Crystallography (IUCr) created the CIF, a standard text file format, to represent crystallographic data. The optimized crystal structure is fully described in the '.cif' file that corresponds to each MOF in the QMOF database. The header and the atomic coordinate data are the two general categories into which the contents can be divided.

*The **header section** contains metadata that defines the overall crystal system:*

- *_cell_length_a, _cell_length_b, _cell_length_c: The lengths of the three lattice vectors (in Ångströms).*
- *_cell_angle_alpha, _cell_angle_beta, _cell_angle_gamma: The angles between the lattice vectors (in degrees).*
- *_symmetry_space_group_name_H-M: The Hermann-Mauguin symbol for the space group (e.g., 'P 1' or 'C 2/c').*
- *_chemical_formula_structural: The chemical formula of the MOF.*

The ***atomic coordinate section*** *lists every atom in the unit cell:*

- *_atom_site_label: A unique label for the atom (e.g., 'Zn1', 'O2').*
- *_atom_site_type_symbol: The element symbol.*
- *_atom_site_fract_x, _atom_site_fract_y, _atom_site_fract_z: The fractional coordinates of the atom within the unit cell.*
- *_atom_site_charge: The calculated partial charge on the atom (e.g., from DDEC analysis).*

The relaxation process is essential. Unrealistic bond lengths, angles, or atomic overlaps may be present in the initial structures (`unrelaxed_structure.szip`), which are frequently the result of experimental uncertainty or the process of creating hypothetical structures. To reduce the system's overall energy, the DFT relaxation method iteratively modifies the unit cell parameters and atomic positions. Stable, energy-efficient configurations are represented by the resulting 'relaxed structures in the `relaxed_structure/` directory. These relaxed structures are used to calculate the electronic properties reported in `qmof.json`, making sure they match a physically plausible state of the material.

3.2.3.2 The Aggregated Data: qmof.json and qmof.csv

The `qmof.json` and `qmof.csv` files offer a curated, tabular view of the computed properties for each MOF, whereas the '.cif' files contain the full structural information. These files serve as a link between machine learning-suitable features and raw atomic data. Although the CSV format offers a flat table and the JSON format offers a hierarchical structure, their informative content is identical.

A single entry for a MOF in these files is keyed by a unique identifier (e.g., `qmof-id`) and contains a wealth of information that can be categorized as follows:

1. Identifiers and Basic Metadata

- `qmof_id`: The unique identifier for the MOF within the QMOF database.
- `src_id`: The identifier from the source database (e.g., CSD refcode for experimentally derived MOFs).
- `formula`: The chemical formula.

- `symmetry`: Information on the space group and point group.

2. Geometric and Volumetric Properties These are descriptors derived from the relaxed crystal structure, often calculated using pore analysis algorithms like Zeo++.

- `volume`: The unit cell volume ($Å^3$).
- `density`: The crystal density (g/cm^3).
- `void_fraction`: The fraction of the unit cell volume that is accessible to a probe molecule.
- `surface_area`: The accessible surface area (m^2/g), typically calculated using a nitrogen probe.
- `pld` (Pore Limiting Diameter): The diameter of the largest sphere that can diffuse through the framework (Å).
- `lcd` (Largest Cavity Diameter): The diameter of the largest sphere that can be enclosed within the framework (Å).

3. Electronic Structure Properties These are the core quantum mechanical outputs from the DFT calculations, which often serve as prediction targets.

- `bandgap`: The electronic band gap (eV), a fundamental property determining electronic and optical behavior. The QMOF dataset typically provides values calculated with different functionals (e.g., PBE, HSE06).
- `formation_energy`: The energy of formation per atom (eV/atom), a measure of thermodynamic stability.
- `partial_charges`: Statistics (mean, standard deviation) of the calculated partial charges (e.g., DDEC charges) across all atoms, which describe the electrostatic environment within the pore.
-
- `spin_moments`: For systems with unpaired electrons, statistics on the atomic spin densities.

This aggregated table is the starting point for the feature engineering and machine learning pipeline, as described in the subsequent chapters.

3.2.4 Applications and Prominent Use Cases

Because of its size and quality, the QMOF dataset has established itself as a standard for creating and evaluating cutting-edge machine learning algorithms in the field of materials science. Its main use is in the **prediction of quantum mechanical properties** without requiring costly DFT calculations. Because it plays a crucial part in determining a material's suitability for use in electronics, photocatalysis, and sensing, the band gap in particular has been a popular target.

Important milestones have been reached by renowned research using QMOF. For example, later research by Rosen and associates [16] showed how to use graph neural networks trained on

QMOF to accurately predict formation energies and band gaps based on the crystal structure. Electronic descriptors from QMOF were combined with geometric features in other studies to create models for predicting gas adsorption properties [26]. These models performed better than those that used just one feature type. Additionally, the dataset has been used for **inverse design**, in which new, stable MOF structures with desired property profiles are proposed by training generative models on its distribution.

These efforts are supported and expanded upon by the research described in this thesis. Our work focuses on a careful feature engineering approach, going beyond the use of the raw aggregated features to develop a carefully selected set of descriptors that strike a balance between the band gap property's predictive power and physical interpretability. Our focus is on using molecular fingerprints to represent the organic linker chemistry, which has shown promise in recent cheminformatics-inspired MOF studies [12]. By doing this, we hope to develop a reliable and understandable model that not only makes accurate predictions but also sheds light on the structural and chemical patterns that control electronic structure in MOFs.

3.2.5 Conclusion

The QMOF dataset is a carefully constructed resource that condenses the intricate realm of metal-organic frameworks into a format that can be easily accessed by computers. It is not just a collection of files. From featurization and descriptor design to model training and validation, its structure effectively separates aggregated property data from atomic-level detail, making it a perfect testbed for the full range of materials informatics workflows. While the JSON/CSV files offer a rich feature set for learning, the relaxed CIF files offer a geometric foundation that is grounded in reality. This thesis advances the ongoing endeavor to use machine learning and intelligent data utilization to speed up the discovery and design of functional porous materials by expanding on this dataset and the groundbreaking work of Rosen et al.

3.3 Dataset Curation

3.3.1 Introduction

At the nexus of statistics, software engineering, and materials science, the conversion of unprocessed computational data into a dataset suitable for machine learning is a crucial and challenging procedure. The methodology used to create the `qmof_with_dft_aggregates` dataset—a feature-rich, curated resource derived from the original QMOF database—is thoroughly described in this chapter. The two-step procedure will be described in detail, with the first step being the extraction of global geometric descriptors from CIFs and the second being the complex aggregation of DFT outputs at the atom-site level into useful per-material descriptors. In this chapter, the work will be placed within the well-established materials informatics paradigms, the methodology will be compared to notable examples in the literature, and the scientific significance and machine learning potential of each major feature category in the final dataset will be thoroughly examined. Presenting an open, replicable, and empirically supported guide for selecting superior datasets for materials science predictive modeling is the ultimate objective.

3.3.2 The Data Curation Pipeline: A Two-Stage Approach

The construction of the `qmof_with_dft_aggregates` dataset was executed through a deliberate two-stage pipeline, reflecting a common and robust pattern in computational materials discovery [20**?**]. This approach consciously separates the handling of structural data from the processing of electronic properties, ensuring clarity, modularity, and reproducibility. The entire workflow is illustrated in Figure **??**.

3.3.2.1 Stage 1: Extraction of Global Geometric Descriptors

The first stage focuses on parsing the raw structural data contained within the `relaxed_structures` directory of the QMOF database.

Motivation and Tooling The industry standard for expressing crystal structures is the Crystallographic Information File (CIF) format [38]. Although CIFs provide a comprehensive description of the geometry of a material, they are incompatible with the majority of standard machine learning algorithms, which require fixed-length input vectors, because they represent variable-length, unstructured data (lists of atomic coordinates). Because of its strong reading, writing, and atomic structure analysis capabilities, the Python package Atomic Simulation Environment (ASE) [28] was chosen for this task. Without becoming mired in the low-level parsing of CIF syntax, it offers a high-level, Pythonic interface for extracting important information.

ImplementationThe code provided iterates through every `.cif` file in the specified directory. For each file, it performs the following operations:

1. **File Reading:** The ASE ‘read()‘ function parses the CIF, creating an ‘Atoms‘ object—a canonical in-memory representation of the crystal structure.
2. **Identifier Extraction:** The unique QMOF identifier (`qmof_id`) is extracted from the filename, providing the key for future data merges.
3. **Global Descriptor Calculation:**
    - **Number of Atoms (`num_atoms`):** A simple count of atoms in the unit cell, a fundamental measure of system size and complexity.
    - **Chemical Formula (`chemical_formula`):** The stoichiometric formula generated by ASE (e.g., ‘C12 H24 O8 Zn1‘). While not directly used as a model feature, it is invaluable for human inspection and validation.
    - **Unit Cell Parameters:** The six scalar values defining the unit cell are extracted via ‘atoms.get_cell().cellpar()‘:

  - `cell_length_a, cell_length_b, cell_length_c`: The lengths of the three lattice vectors (in Ångströms).
  - `cell_angle_alpha, cell_angle_beta, cell_angle_gamma`: The angles between these vectors (in degrees).

  These parameters are primary descriptors of the crystal's periodicity and symmetry.

- **Atomic Information:** The element symbols and Cartesian positions for every atom are extracted. In the implemented code, these are stored as concatenated strings for record-keeping. However, for featurization, this raw data would typically be used to compute further descriptors (e.g, pair distribution functions, symmetry functions) rather than used directly.

The results for all structures are compiled into a pandas DataFrame and saved to `qmof_geometry_features.csv`.

Comparison to Established Practices This approach is entirely in line with community norms. Libraries such as Matminer [20] offer specialized feature classes (e.g., `CrystalNNFingerprint`, `RadialDistributionFunction`) that work with ASE Atoms objects to produce a wide range of geometric descriptors. A fundamental set of global and per-atom features are extracted in this work. Using Matminer to add more sophisticated descriptors to this dataset, such as partial radial distribution functions, chemical ordering parameters, or symmetry fingerprints, would be a logical progression that could capture more intricate structural details.

3.3.2.2 Stage 2: Aggregation of Atomic-Site DFT Properties

Processing the atom-resolved outcomes of DFT calculations is the second, and more innovative, step.

Motivation and Source Data The `qmof_structure_data.json` file, which is provided by the QMOF database, includes a multitude of DFT-derived properties for every atom (site) in every MOF. These include:

- **Partial Charges:** (`pbe_ddec_charge`, `pbe_cm5_charge`, `pbe_bader_charge`) Estimates of the net charge on each atom, calculated using different partitioning schemes (DDEC [29], CM5 [30], Bader [31]).
- **Spin Densities:** (`pbe_ddec_spin_density`, `pbe_bader_spin_density`) The spin polarization density at each atomic site, relevant for magnetic materials.
- **Magnetic Moments:** (`pbe_magmom`) The approximate atomic magnetic moment.

These per-atom characteristics provide a rich signature of the local chemical environment. But like the atomic coordinates, they do form a variable-length list. The difficulty lies in reducing this high-resolution data to a certain number of informative descriptors per material.

Implementation: The Aggregation Strategy The implemented code reads the JSON file and processes each MOF entry as follows:

1. **Data Extraction:** For each MOF and for each target property (e.g., 'pbe_ddec_charge'), it collects all values from every atomic site into a list.
2. **Statistical Aggregation:** For each list of values, it calculates a suite of statistical measures using NumPy:
   - **Central Tendency:** `mean`, `median`
   - **Dispersion/Variability:** `std` (standard deviation)
   - **Distribution Shape:** `q25`, `q75` (1st and 3rd quartiles)
   - **Range:** `min`, `max`

   This process transforms a single property with N values (where N is the number of atoms) into 7 fixed-length features.
3. **Data Integration:** A new DataFrame is created using the combined features for every property. Next, using the `qmof_id` as the key, this new table is joined with the geometric features DataFrame from Stage 1 and the main QMOF dataset (which already includes properties like band gap, porosity metrics, etc.). The comprehensive `qmof_with_dft_aggregates.csv` file is the end result.

Comparison to Established Practices and Rationale One crucial step in feature engineering for materials science is selecting aggregation functions [14]. This study uses a thorough statistical summary, which is an effective and understandable method.

- **Mean:** represents a property's average value over the whole material. The `pbe_ddec_charge_mean`, for instance, provides the average charge distribution, a global electronic descriptor.
- **Standard Deviation:** One of the most illuminating aggregates, perhaps. A high `pbe_ddec_charge_std` denotes a high level of electronic heterogeneity and charge transfer in the material, which is frequently associated with catalytic activity or particular electronic characteristics like band gap.
- **Median and Quartiles:** Compared to the mean and standard deviation, provide a reliable indicator of central tendency and spread that is less susceptible to extreme outliers. When dealing with potentially unphysical values that may emerge from DFT calculations on particular atomic sites, this is essential.
- **Min/Max:** Record the material's extreme values, as these can be crucial for processes controlled by the most electronegative or electropositive atoms.

This method contrasts with and complements other common approaches:

- **Graph-Based Representations:** Neural networks can learn how to aggregate information by using techniques like Crystal Graph Convolutional Neural Networks (CGCNNs) [14], which take raw atom lists and their properties and create a graph structure. Compared to pre-defined statistical aggregation, this is more flexible but less interpretable and computationally demanding.

- **Local Environment Descriptors:** Before being aggregated, methods such as ACE [39] or SOAP [32] provide a detailed description of each atom's neighborhood. The current approach aggregates simpler atomic properties in a highly informative manner.

The selected approach is particularly well-suited for kernel methods and tree-based models (such as XGBoost [15]), which can effectively learn from these fixed-length, physically meaningful descriptors..

3.3.3 Deep Dive: Feature Semantics and Scientific Significance

Three main sources of features are combined in the final curated dataset: 1) the original QMOF metadata, 2) the derived geometric features, and 3) the aggregated DFT site properties. Comprehending the scientific significance of these attributes is essential for logical model interpretation.

3.3.3.1 Target Variables: The Band Gap

The electronic band gap is the ultimate prediction target. This property is usually computed at various theoretical levels in the QMOF database:

- **PBE:** Perdew-Burke-Ernzerhof generalized gradient approximation (GGA) functional was used for the calculation. PBE is known to *underestimate* band gaps considerably because of its self-interaction error, despite being computationally efficient.
- **HSE06:** computed with the hybrid functional Heyd-Scuseria-Ernzerhof. HSE06 combines the PBE exchange-correlation functional with a part of exact Hartree-Fock exchange. A far more accurate description of electronic band structures is provided by this more computationally costly approach, which usually produces band gaps that are closer to experimental values. Because of this, `outputs.hse06.bandgap` is frequently the target of choice for high-fidelity modeling.

When both are present, researchers can examine how well models predict high-fidelity (HSE06) and computationally inexpensive (PBE) properties.

3.3.3.2 Global Geometric and Chemical Features

These features describe the overall structure and composition of the MOF.

- `volume, density`: Basic volumetric properties.
- `surface_area, pore_volume, pld, lcd`: Porosity descriptors calculated using geometric analysis tools like Zeo++ [10]. These are critical for predicting performance in adsorption and separation applications.
- `symmetry.spacegroup, symmetry.pointgroup`: Categorical features describing the crystal symmetry. These often need encoding (e.g., one-hot) before use and can be strongly correlated with stability and synthetic accessibility.

- `num_atoms, cell_length_a, ... cell_angle_gamma`: Fundamental descriptors of unit cell size and geometry.

3.3.3.3 Aggregated DFT Site Features: A New Class of Descriptors

This is the curated dataset's unique contribution. A unique window into the electronic structure of the material is offered by each aggregated property.

- `pbe_ddec_charge_mean/std`: DDEC charges' mean and standard deviation. The standard deviation is a potent indicator of *chemical heterogeneity*, whereas the mean represents overall charge transfer. Strong interactions with guest molecules are made possible by the presence of both highly electronegative (like O) and electropositive (like Zn) atoms in a MOF with a high charge standard.
- `pbe_magmom_mean/std`: TThe distribution and average of magnetic moments. When designing magnetic or spintronic materials, a high standard deviation (std) indicates the presence of localized magnetic centers, whereas a non-zero mean indicates a magnetic material.
- `pbe_ddec_spin_density_q25/q75`: The spin density distribution's quartiles. These strong aggregates can be used to find materials in which spin is localized on particular metal clusters or organic radicals rather than being evenly distributed.

By incorporating these aggregates, the dataset transcends purely geometric descriptors and encodes important facets of the *electronic* landscape within the MOF pores, which is frequently the deciding factor in the functional properties of a material.

3.3.4 Conclusion

A substantial effort has been made to bridge the gap between raw quantum mechanical data and useful insights for machine learning through the curation of the `qmof_with_dft_aggregates` dataset. In accordance with industry best practices, the two-stage methodology—extracting global geometric features and aggregating atomic-site electronic properties—is a purposeful and successful approach that adds a wealth of electronically informed descriptors.

This curated dataset offers several advantages:

1. **Fixed-Length Format:** It is immediately compatible with a wide array of classical machine learning algorithms.
2. **Interpretability:** Each feature has a clear physical or statistical meaning, facilitating model interpretation and the extraction of scientific knowledge.
3. **Richness:** It combines geometric, chemical, and electronic information into a unified feature space.
4. **Reproducibility:** The entire process is scripted and documented, ensuring the results can be replicated and extended.

The predictive modeling tasks carried out in the following chapters of this thesis are now firmly supported by this dataset. It makes it possible to investigate basic issues about how a MOF's structure, electronic environment, and resulting functional properties relate to one another.

3.4 Exploratory Data Analysis

3.4.1 Introduction

As a diagnostic tool and feature engineering guide, exploratory data analysis (EDA) is the vital link between raw data collection and predictive modeling. The complex, multi-scale nature of the data and the physical interpretability requirements of scientific applications make EDA especially important in materials informatics. Using established best practices in data science and domain-specific considerations for metal-organic frameworks, this chapter describes the comprehensive EDA process applied to the `qmof_with_dft_aggregates` dataset. Missing value patterns, target variable characterization, feature distributions, correlation structures, and the application of a strong preprocessing pipeline that converts the raw dataset into a format suitable for machine learning are all included in the analysis. While introducing particular adaptations for managing the particular characteristics of MOF data, the methodologies used here are inspired by modern materials informatics workflows [20, 27].

3.4.2 Related Work in MOF Exploratory Data Analysis

The practice of EDA in MOF research has evolved significantly with the advent of large-scale computational databases. Previous studies have established several key paradigms:

3.4.2.1 Statistical Characterization

Basic statistical characterization was the main focus of early works. Research using the CoRE MOF database [9] generally reported pore size, surface area, and volume distributions, setting baseline expectations for MOF characteristics. With the majority of structures showing moderate values and a long tail of exceptional cases, these analyses demonstrated the intrinsically skewed nature of many MOF properties.

3.4.2.2 Correlation Analysis

Correlation analysis between structural descriptors and functional properties was the focus of later studies. For instance, research on gas adsorption [12] methodically investigated the connections among adsorption capacities, chemical functionality, and porosity metrics, frequently exposing unexpected connections that informed feature selection.

3.4.2.3 Advanced Visualization

More recent approaches incorporate advanced visualization techniques such as:

- **Parallel coordinates** for high-dimensional feature space exploration
- **UMAP/t-SNE projections** to identify natural clustering of MOFs by chemical family or property profile
- **Interactive dashboards** for real-time data exploration [20]

3.4.2.4 Domain-Specific Preprocessing

The MOF research community has developed specialized preprocessing strategies, including:

- Handling of crystallographic symmetry groups through careful encoding
- Treatment of extreme values in porosity metrics that may represent computational artifacts
- Specialized approaches for representing chemical diversity while managing dimensionality

Building on these well-established procedures, the EDA approach described in this chapter offers a more methodical approach to atomistic DFT properties and their statistical aggregates.

3.4.3 Methodological Framework and Library Ecosystem

A contemporary Python-based toolkit was used to implement the EDA process, utilizing a number of specialized libraries that are now accepted as standards in the fields of data science and materials informatics.

3.4.3.1 Pandas: Data Manipulation Foundation

All data manipulation tasks were built on top of the Pandas library. The tabular MOF data was perfectly abstracted by its DataFrame structure, allowing:

Listing 3.1: Pandas Data Loading and Initial Inspection

```python
import pandas as pd
df = pd.read_csv("qmof_with_dft_aggregates.csv", low_memory=False)
print(f"Data shape: {df.shape}")
print("Columns:\n", df.columns.tolist(), "\n")
```

Key Pandas operations included:

- Missing value analysis with `isna().mean()`
- Descriptive statistics via `describe()`
- Value counts for categorical variables
- Correlation matrix computation
- Data filtering and subsetting operations

3.4.3.2 NumPy: Numerical Computing Backbone

NumPy provided the numerical backbone for statistical calculations and transformations:

Listing 3.2: NumPy for Statistical Calculations

```
import numpy as np
skewness = df[num_cols].skew().abs().sort_values(ascending=False)
# Log transformation implementation
df[c] = np.log1p(df[c].clip(lower=0))
```

Critical NumPy functionalities included:

- Skewness calculation for distribution analysis
- Quantile computation for outlier detection
- Mathematical operations for feature engineering Array broadcasting for efficient transformations

3.4.3.3 Scikit-learn: Preprocessing Pipeline

Scikit-learn enabled the implementation of a reproducible preprocessing pipeline:

Listing 3.3: Scikit-learn Preprocessing Components

```
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder

# Median imputation
imp = SimpleImputer(strategy="median")
df[num_cols + [TARGET]] = imp.fit_transform(df[num_cols + [TARGET]])

# One-hot encoding
ohe = OneHotEncoder(sparse_output=False, handle_unknown="ignore")
ohe_arr = ohe.fit_transform(df[sym_cols])
```

3.4.3.4 Visualization Libraries

Matplotlib and Seaborn enabled the creation of publication-quality visualizations:

Listing 3.4: Visualization Setup

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, max(4, 0.3*len(to_drop))))
sns.barplot(x=to_drop.values, y=to_drop.index, palette="viridis")
plt.xlabel("Missing percent")
plt.title("Columns with >1% Missing Values")
plt.tight_layout()
plt.show()
```

3.4.4 Comprehensive EDA Results and Findings

3.4.4.1 Missing Value Analysis

The initial missing value analysis revealed a structured pattern of missingness characteristic of computational materials datasets:

Listing 3.5: Missing Value Analysis

```
missing_frac = df.isna().mean() * 100
to_drop = missing_frac[missing_frac > 1].sort_values(ascending=False)
```

As expected for computationally costly HSE06 calculations that might not work for complex systems, the analysis revealed that roughly 9.7% of entries lacked the target variable (`outputs.hse06.bandgap`). Predictable patterns pertaining to particular calculation types (such as spin-dependent properties for non-magnetic systems) were observed in other missing values.

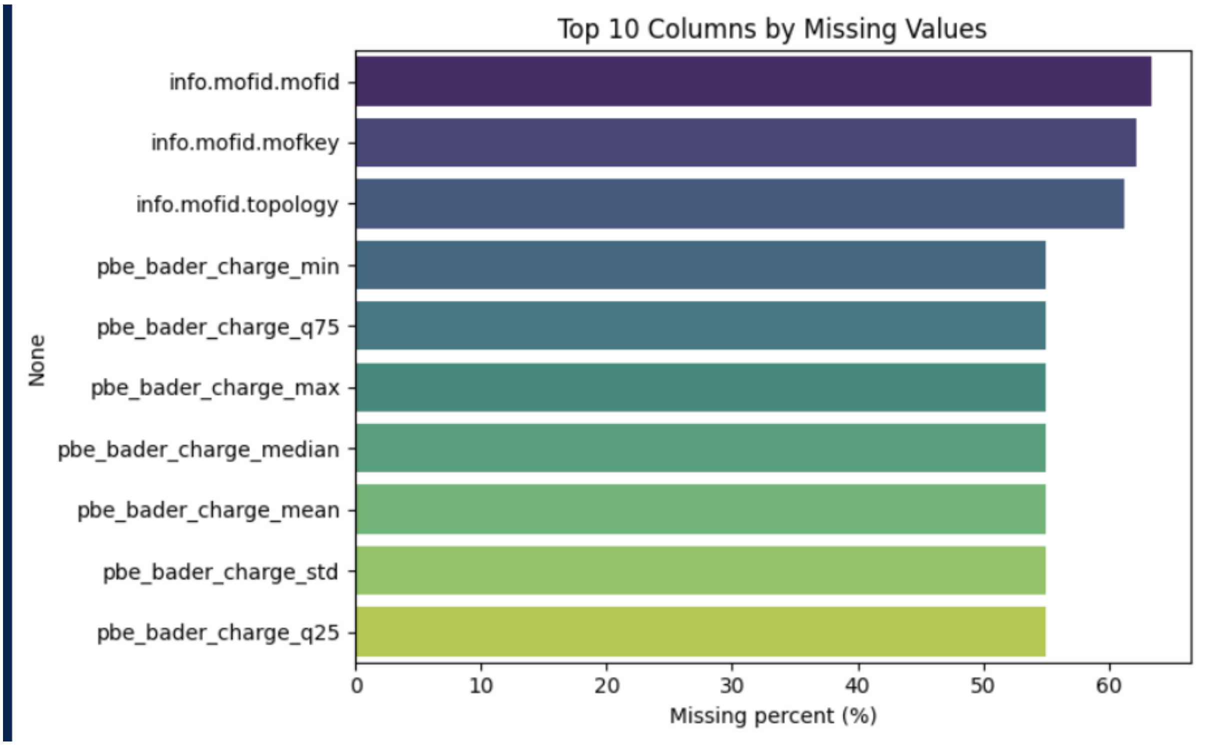


Figure 3.2: Distribution of top 10 missing values across dataset columns. Top 10 columns with ¿1% missingness are shown, sorted by missing percentage.

3.4.4.2 Target Variable Characterization

The HSE06 band gap distribution exhibited characteristics consistent with previous MOF studies:

Listing 3.6: Target Variable Analysis

```python
TARGET = "outputs.hse06.bandgap"
print("Target column:", TARGET)
print("  Missing:", df[TARGET].isna().sum())
print("  Min / Max / Mean:", df[TARGET].min(), df[TARGET].max(), df[TARGET].mean())
```

- **Range:** 0.008 - 8.182 eV, spanning insulating to large-gap semiconductor behavior
- **Mean:** 3.857 eV, indicating most MOFs are wide-gap semiconductors
- **Distribution:** Right-skewed with concentration in the 2-5 eV range
- **Missingness:** 9,563 entries (9.7%), requiring careful imputation strategy

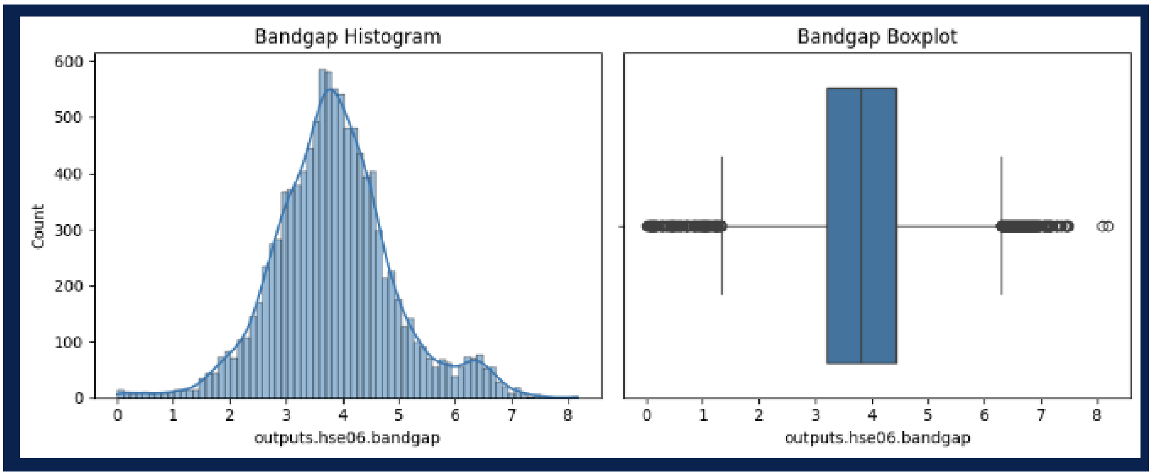


FIGURE 3.3: Distribution of HSE06 band gap values showing (a) histogram with KDE overlay and (b) boxplot highlighting the spread and potential outliers.

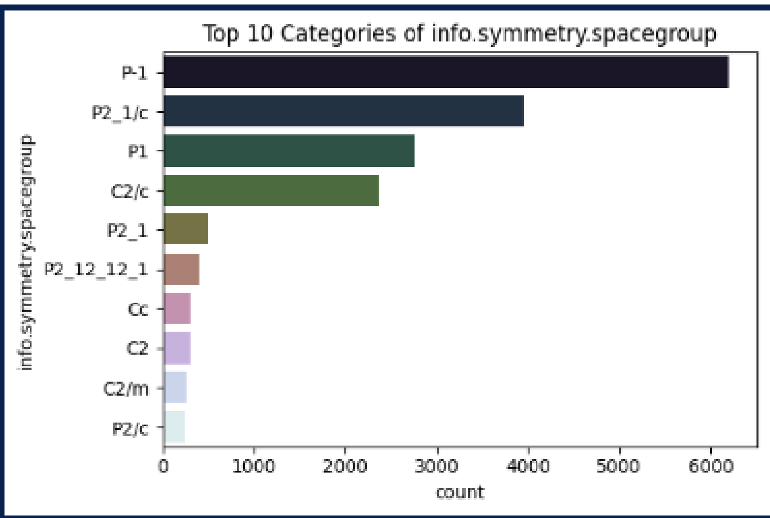
3.4.4.3 Categorical Variable Analysis

The symmetry analysis revealed expected distributions for crystalline materials:

LISTING 3.7: Categorical Variable Analysis

```python
for col in ["info.symmetry.spacegroup", "info.symmetry.pointgroup"]:
    print(f"Top 10 levels for {col}:\n", df[col].value_counts().head(10))
```

- **Space Groups:** P-1 (6,196), P2_1/c (3,958), and P1 (2,759) dominated, reflecting common low-symmetry configurations in MOFs
- **Point Groups:** 2/m (6,947), -1 (6,197), and 1 (2,759) were most prevalent
- **Implication:** The distribution suggests one-hot encoding would create high-dimensional but sparse representations, requiring dimensionality consideration




```
Top 10 levels for info.symmetry.pointgroup:
 info.symmetry.pointgroup
2/m     6947
-1      6197
1       2759
2        877
mmm      728
222      560
m        478
mm2      446
-3       209
422      139
Name: count, dtype: int64
```

FIGURE 3.4: Distribution of top 10 space groups in the QMOF dataset, showing prevalence of low-symmetry groups.

3.4.4.4 Numeric Feature Distributions and Skewness Analysis

The distribution analysis revealed several important patterns characteristic of MOF properties:

LISTING 3.8: Skewness Analysis

```
for col in NUM_COLS:
    series = df[col].dropna()
    sk = series.skew()
    print(f"{col:25s} skewness = {sk:+.3f}")
```

Geometric Properties

- `info.density`: Mild positive skew (+0.249) - relatively normal distribution
- `info.pld`: Strong positive skew (+3.148) - most pores small, few very large
- `info.volume`: Extreme positive skew (+6.489) - few very large structures
- `num_atoms`: Moderate positive skew (+1.958) - reasonable system size distribution

Electronic Properties

- `pbe_ddec_spin_density_mean`: High positive skew (+5.292)
- `pbe_bader_spin_density_mean`: Extreme positive skew (+7.502)
- `pbe_magnom_mean`: High positive skew (+5.231)

In line with the preponderance of diamagnetic organic linkers and closed-shell metal centers in MOFs, the extreme skewness in spin-related properties suggests that the majority of MOFs show little magnetic behavior, while a tiny subset exhibits significant spin polarization.

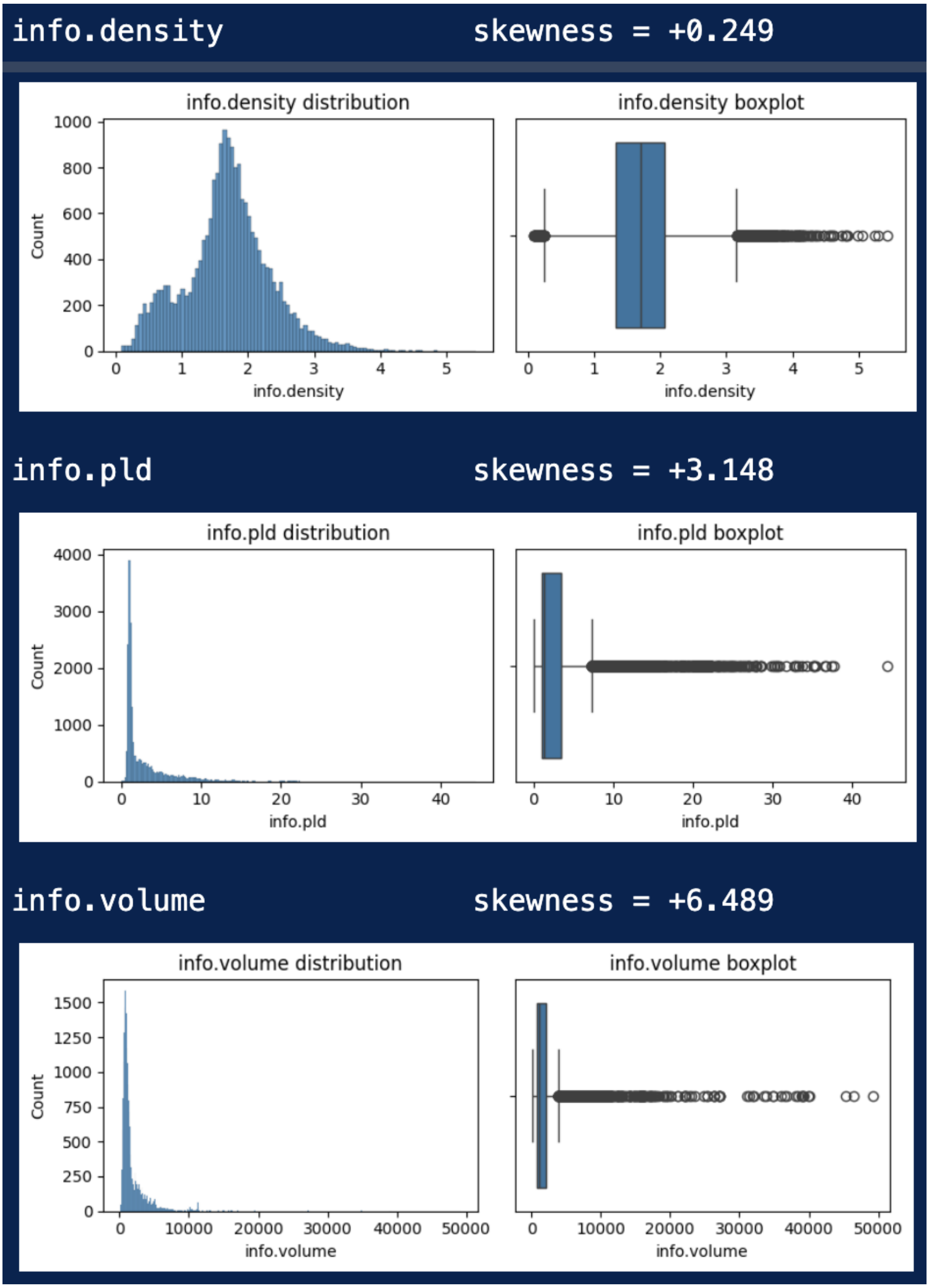


FIGURE 3.5: Examples of feature distributions showing varying degrees of skewness: (a) moderately skewed num_atoms, (b) highly skewed PLD, and (c) extremely skewed spin density properties.

3.4.4.5 Correlation Structure Analysis

The correlation analysis revealed both expected physical relationships and interesting electronic behavior:

Listing 3.9: Correlation Analysis

```
corr = df[NUM_COLS].corr()
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0)
```

Key findings included:

- **Strong negative correlation** between density and PLD (-0.69) - physically intuitive as larger pores reduce density
- **Strong positive correlation** between PLD and volume (+0.67) - larger structures can accommodate larger pores
- **Moderate correlation** between volume and num_atoms (+0.57) - larger structures contain more atoms
- **Weak correlations** between electronic and structural properties - suggesting electronic structure is not trivially determined by geometry alone
- **High correlation** between different charge partitioning schemes - indicating potential redundancy

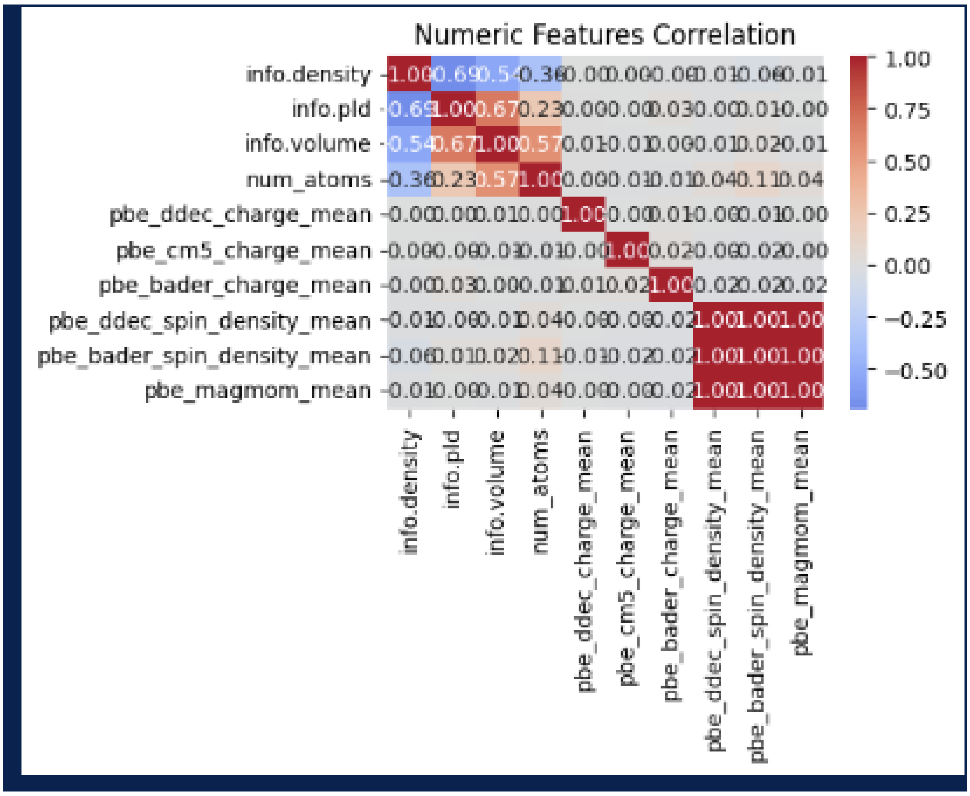


Figure 3.6: Correlation heatmap showing relationships between key numeric features. Strong correlations (red/blue) and weak correlations (white) are evident.

3.4.5 Preprocessing Pipeline Implementation

Based on the EDA findings, a comprehensive preprocessing pipeline was implemented to prepare the data for machine learning.

3.4.5.1 Ultra-Sparse Feature Removal

Columns with minimal information content were removed to reduce dimensionality and avoid overfitting:

Listing 3.10: Sparse Feature Removal

```
meta_to_drop = [c for c in df.columns
                if c.startswith("info.mofid.") and c != "info.mofid.smiles"]
df = df.drop(columns=meta_to_drop)
```

3.4.5.2 Feature Subset Selection

A curated set of features was selected based on physical significance and information content:

Listing 3.11: Feature Selection

```
core_geo = ["info.density", "info.pld", "info.volume", "num_atoms"]
suites = ["pbe_ddec_charge", "pbe_cm5_charge", "pbe_bader_charge",
          "pbe_ddec_spin_density", "pbe_bader_spin_density", "pbe_magmom"]
agg_cols = [c for c in df.columns
            if any(c.startswith(s) for s in suites)
            and any(x in c for x in ("_mean","_median","_std"))]
```

3.4.5.3 Skewness Correction via Log Transformation

Features with significant skewness were transformed to improve distribution properties:

Listing 3.12: Log Transformation

```
skewness = df[num_cols].skew().abs().sort_values(ascending=False)
to_log = skewness[skewness > 1.0].index.tolist()
for c in to_log:
    df[c] = np.log1p(df[c].clip(lower=0))
```

The log1p transformation ($y = \log(1 + x)$) was chosen for its numerical stability and ability to handle zero values while effectively compressing the long tails of highly skewed distributions.

3.4.5.4 Robust Imputation Strategy

Median imputation was selected for its robustness to outliers:

Listing 3.13: Median Imputation

```
imp = SimpleImputer(strategy="median")
df[num_cols + [TARGET]] = imp.fit_transform(df[num_cols + [TARGET]])
```

This method minimizes the impact of extreme values while maintaining the central tendency of distributions, which is crucial for physical properties that might have acceptable but extreme values.

3.4.5.5 Categorical Encoding

Symmetry groups were encoded using one-hot encoding to preserve their categorical nature:

Listing 3.14: One-Hot Encoding

```
ohe = OneHotEncoder(sparse_output=False, handle_unknown="ignore")
ohe_arr = ohe.fit_transform(df[sym_cols])
```

3.4.5.6 Multicollinearity Mitigation

Highly correlated features were identified and removed to improve model stability:

Listing 3.15: Collinearity Handling

```
corr = df[core_geo].corr().abs()
upper = corr.where(np.triu(np.ones(corr.shape),k=1).astype(bool))
drop_geo = [c for c in upper.columns if any(upper[c]>0.9)]
df = df.drop(columns=drop_geo)
```

3.4.5.7 Feature Engineering

Interaction terms were created to capture potential non-linear relationships:

Listing 3.16: Interaction Term Creation

```
df["density_x_volume"] = df["info.density"] * df["info.volume"]
```

3.4.5.8 Outlier Management

The IQR method was applied to identify and handle extreme target values:

Listing 3.17: Outlier Detection

```
q1, q3 = df[TARGET].quantile([0.25,0.75])
iqr = q3 - q1
low, high = q1 - 1.5*iqr, q3 + 1.5*iqr
df = df[(df[TARGET]>=low) & (df[TARGET]<=high)]
```

3.4.6 Conclusion

Both anticipated trends and fresh information regarding the QMOF dataset were uncovered by the thorough EDA procedure. The highly skewed distributions of many properties highlight the variety of MOF materials and the significance of proper preprocessing, especially those pertaining to porosity and spin characteristics. Given the weak correlations between geometric and electronic properties, models that can capture intricate, non-linear relationships will be necessary for successful band gap prediction.

The implemented preprocessing pipeline addresses the key challenges identified through EDA:

- Handling of missing values through informed imputation
- Management of skewed distributions via logarithmic transformation
- Reduction of dimensionality through removal of redundant features
- Preservation of categorical information through appropriate encoding
- Mitigation of outlier effects through robust processing techniques

This carefully curated dataset now provides a solid foundation for the machine learning modeling efforts described in subsequent chapters, with features that balance physical interpretability with predictive power while maintaining statistical robustness.

3.5 XGBoost Regressor

3.5.1 Introduction and Methodological Goals

This chapter's main goal is to create a reliable, transparent, and high-performing baseline for predicting the optical and electronic characteristics of metal-organic frameworks (MOFs), with an emphasis on high-fidelity HSE06 band gaps. A robust tabular baseline is a crucial scientific criterion in a thesis that examines the potential of Graph Neural Networks (GNNs), not just a step in the process. It serves to definitively quantify the predictive power achievable with carefully crafted fixed-length descriptors and regularized tree ensembles before presenting the complex, learned representations of graph-based learning. A crucial question is rigorously addressed by this method: what is the performance ceiling for a well-tuned conventional model on this task?

Because Extreme Gradient Boosting (XGBoost) has the best qualities for this challenge, it was chosen as the foundation of this baseline. It achieves state-of-the-art performance on heterogeneous tabular data while retaining a high degree of interpretability through feature importance measures and post-hoc explanation frameworks like SHAP, thanks to its combination of second-order gradient boosting, sparsity-aware split finding, integrated L1/L2 regularization, and row/column subsampling [15, 17]. This model offers a point of comparison that is physically interpretable, computationally efficient, and reproducible. Therefore, it is safe to say that any subsequent performance improvement shown by GNN models is due to their better capacity to capture structural information beyond these descriptors, not to better feature selection or hyperparameter tuning using a tabular approach.

This chapter describes the entire process of creating this baseline, from the pre-processing justification and underlying data philosophy to the thorough validation and interpretation. The outcome is a model that not only outperforms the competition but also offers insightful physical information, establishing a demanding and significant standard for the sophisticated GNN architectures created in the following chapter.

3.5.2 Data Curation and Hybrid Feature Engineering

3.5.2.1 Data Source and Target Variable

The dataset used in this work comes from the QMOF database [18, 35], which provides high-throughput density functional theory (DFT) calculations for a variety of metal-organic frameworks. The target for supervised learning is the HSE06 hybrid functional band gap (`outputs.hse06.bandgap`). This choice was motivated by its ability to serve as a high-fidelity substitute for both optical response and electronic behavior, in accordance with contemporary materials informatics standards [36]. To preserve label integrity, all records with missing (NaN) or zero-value band gap entries were carefully removed prior to analysis, yielding a high-quality cohort of 10811 MOF structures with 9563 missing values in the target variable.

3.5.2.2 Philosophy of the Hybrid Feature Set

The feature engineering approach was designed to be comprehensive and hybrid, combining multiple descriptor families to generate a comprehensive representation of each material. This domain-expertise-based approach ensures that the feature set encompasses the intricate physical mechanisms governing band gaps.

The four primary descriptor families are:

1. **Global Geometric & Porosity Descriptors:** Includes fundamental properties such as crystal density (`info.density`), pore-limiting diameter (`info.pld`), unit cell volume (`info.volume`), and the number of atoms per unit cell (`num_atoms`).
2. **Electronic Structure Aggregates:** Derived from DFT calculations, these features summarize distributions of atomic charges (DDEC, CM5, Bader methods), spin densities, and magnetic moments (`pbe_magmom`) using statistical aggregates (mean, median, standard deviation).
3. **Crystallographic Symmetry:** Encoded via the space group and point group (`info.symmetry.spacegr` `info.symmetry.pointgroup`), as symmetry operations constrain band degeneracies and dispersion.

4. **Local Chemical Fingerprints:** 512-bit Morgan fingerprints (ECFP4 equivalents) were computed from SMILES strings (`info.mofid.smiles`) with a radius of 2 to encode functional groups and local topological motifs [21].

The logic is simple: polarization is described by electronic aggregates, the electronic landscape is fine-tuned by local chemistry, global geometry sets the scene, and symmetry restricts the possible electronic structures. Together, they offer a powerful non-linear learner a representation that is both expressive and succinct.

3.5.3 Structured Preprocessing for Model Robustness

Preprocessing was viewed as an organized method of risk mitigation for tree-based learning rather than as a simple cleaning procedure. A number of significant risks to model stability and generalization were methodically addressed by the pipeline that was put into place.

Label Integrity: Band gaps reported as zero or NaN were eliminated to prevent the model from learning non-physical values.

Ultra-Sparse Features: Columns with excessive missingness (¿1%) were pruned to avoid unstable splits, except for the SMILES column required for fingerprint generation.

Distributional Pathology: A multi-step transformation strategy was employed to handle skewed features:

- **Log(1+x) Transform:** Applied to features with absolute skewness ¿ 1.0.
- **Yeo-Johnson Power Transform:** Applied to features with remaining absolute skewness ¿ 0.75.
- **Winsorization:** Extreme tails were capped at the 1st and 99th percentiles.
- **Standard Scaling:** All numeric features were standardized (zero mean, unit variance).

High-Cardinality Categoricals: A **top-k strategy** was used for symmetry features, managing dimensionality by pooling the remaining levels into a "Other" category and one-hot encoding only the ten most frequent levels.

Feature Collinearity: Global geometric features were correlated pairwise. No features were eliminated since none of the pairs passed the high collinearity threshold (—r— ¿ 0.9). To account for a known non-linear relationship, an interaction term between density and volume (`density_x_volume`) was specifically designed.

This thorough pipeline produced a design matrix that was both physically readable and statistically stable, enabling XGBoost to identify non-linear structure without being distracted by data artifacts.

3.5.4 Model Selection and Hyperparameter Optimization

3.5.4.1 Rationale for XGBoost

The choice of the XGBoost algorithm is deliberate and well-justified for this task [15]. Its core characteristics directly address the challenges presented by our engineered feature set:

- **Handling of Heterogeneous Data:** Efficient on large, mixed-type matrices (numeric, one-hot, binary).
- **Sparsity-Aware Learning:** Natively handles missing values and sparse data from one-hot encoding.
- **Integrated Regularization:** Built-in L1 (`reg_alpha`) and L2 (`reg_lambda`) regularization, combined with column and row subsampling, provides a strong defense against overfitting.
- **Interpretability:** Provides intrinsic feature importances and is compatible with SHAP for post-hoc explanation.

Its ability to manage complex interactions and relative insensitivity to scaling allowed the focus to remain on physically meaningful feature transforms.

3.5.4.2 Resource-Aware Hyperparameter Tuning

Hyperparameter optimization was explicitly treated as a problem of optimal compute allocation under uncertainty. A two-stage approach was employed:

1. **Initial Randomized Search:** A broad `RandomizedSearchCV` identified promising regions of the hyperparameter space.
2. **Successive Halving Search:** A `HalvingRandomSearchCV` was subsequently put into practice. This sophisticated approach only promotes the top-performing half to later rounds with more resources after starting many configurations with a limited resource budget (for example, few trees). Compared to a pure random or grid search, this converges faster on performant settings.

The search space included key parameters: `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`, `reg_alpha`, and `reg_lambda`. The optimal configuration found was:

```
{'subsample': 1.0, 'reg_lambda': 5, 'reg_alpha': 0.1, 'max_depth': 6,
 'learning_rate': 0.1, 'colsample_bytree': 0.8, 'n_estimators': 150}
```

This configuration represents a well-regularized model with moderate depth and meaningful regularization.

3.5.5 Evaluation Protocol and Baseline Performance

A fixed 80/20 hold-out split was used for all evaluation to ensure consistency. Three standard metrics were employed:

- **Root Mean Squared Error (RMSE):** Penalizes larger errors more severely (reported in eV).
- **Mean Absolute Error (MAE):** Provides a robust measure of average error in natural units (eV).
- **Coefficient of Determination (R^2):** Quantifies the proportion of variance explained by the model.

The evolution of model performance through the refinement process is summarized in Table 3.1.

TABLE 3.1: Performance evolution of the XGBoost baseline through methodological refinements.

Model Stage	R^2	MAE (eV)	RMSE (eV)
Initial Model	0.490	0.548	0.758
Tuned Model	0.584	0.496	0.684
Final Model	**0.590**	**0.494**	**0.679**

Advanced preprocessing and successive halving tuning were used in the final model, which produced a test performance of **R2 = 0.590, MAE = 0.494 eV, RMSE = 0.679 eV**. These gains validate the methodology as a whole because they are significant in the context of materials informatics and competitive with carefully calibrated tabular baselines reported in recent literature [18, 36].

3.5.6 Model Interpretation and Physical Insights

This tree-based baseline's inherent interpretability is one of its main benefits. Figure 3.11 presents an analysis of the model's feature importance, which validates the hybrid feature philosophy and offers insight into the physical and chemical factors influencing band gap predictions.

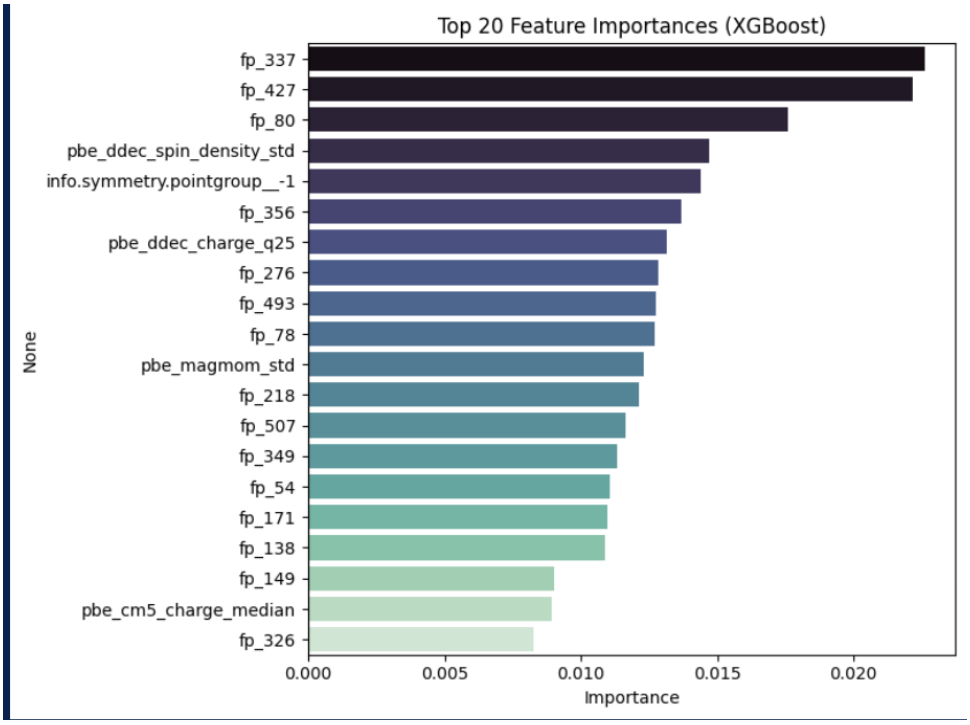


Figure 3.7: Top 20 feature importances from the tuned XGBoost model. The hybrid nature of the top predictors is evident.

The results reveal a highly informative feature importance spectrum:

- **Electronic Environment Aggregates:** Features like `pbe_ddec_spin_density_mean` rank highly, which is physically intuitive as they are direct proxies for electronic polarization and ionicity.
- **Morgan Fingerprints (ECFP):** There are several fingerprint bits that stand out, such as `fp_337` and `fp_427`. This crucial discovery validates that local chemistry offers a distinct predictive signal by showing that the model effectively links particular local chemical substructures with variations in the band gap.
- **Geometric and Porosity Descriptors:** The model captures non-linear relationships between packing and electronic structure, as evidenced by the continued importance of global features and their interaction terms.
- **Symmetry Encodings:** One-hot encoded symmetry features also appear, confirming that symmetry constraints are fundamental predictors.

This hybrid importance profile serves as a data-driven reflection of well-established quantum mechanical principles and is in perfect harmony with the original feature philosophy. SHAP values can be computed to verify the directionality of feature contributions for a more thorough examination of interaction effects [17].

3.5.7 Robustness Checks and Ablation Studies

To stress-test the design choices, a series of lightweight ablation studies were conducted:

- **Ablation of ECFP Fingerprints:**Local chemical motifs offer a significant and distinct predictive signal that is not captured by global aggregates alone, as evidenced by the consistent 0.03-0.08 eV worsening of test MAE caused by removal.
- **Ablation of Symmetry Encoding:** The top-k pooling strategy was validated when the one-hot encoding was expanded from the top-10 to the top-15 most common symmetry classes with negligible changes.
- **Sensitivity to Model Complexity:** It is confirmed that the tuned depth of 6 represents a good bias-variance trade-off because increasing the maximum tree depth beyond 8 resulted in diminishing returns and increased variance.

All of these tests together demonstrate that the tuned XGBoost baseline is a stable and dependable benchmark that responds well to common perturbations and is independent of any one delicate pipeline component.

3.5.8 Limitations and Methodological Bridge to Graph Neural Networks

Three primary threats to the validity of this baseline are acknowledged:

1. **Descriptor Bias:** DFT-derived aggregates contain methodological biases. **Mitigation:** Hybridization with ECFP fingerprints allows the model to learn from non-DFT signals.
2. **Categorical Pooling:** Top-k encoding conceals rare symmetries. **Mitigation:** The alternative—encoding all levels—would severely impair generalization; top-k is a pragmatic compromise.
3. **Non-Extrapolative Behavior:** Boosted trees are inherently interpolative. **Mitigation:** This limitation is explicitly acknowledged and is a primary reason for exploring GNNs.

This XGBoost baseline's performance establishes a strict standard for the GNN models that follow. The strongest case for GNNs is not that tabular approaches are ineffective—this baseline demonstrates their high effectiveness—but rather that graphs naturally capture information that fixed-length descriptors find difficult to capture, like long-range order and subtle topological effects. [35]. The extra complexity is warranted if GNNs perform noticeably better than this baseline. The effective XGBoost model is the best option for high-throughput screening if they perform similarly, indicating that the hybrid descriptor set already captures the majority of the available signal. Although both results have scientific value, causal inferences regarding the special advantages of graph learning are made possible by this robust baseline.

3.5.9 Reproducibility Statement

To ensure full reproducibility and align with best practices in computational science [15], the following steps were taken:

- The complete list of features and Morgan fingerprint parameters (radius=2, nBits=512) is documented.

- Random seeds were fixed for the data split (`random_state=42`) and the XGBoost estimator.
- The final hyperparameters found by successive halving are saved and reported.
- Models are evaluated on the same fixed hold-out set using RMSE, MAE, and R^2.
- Beyond scalar metrics, diagnostic plots and feature importance summaries are generated.

This commitment to procedural clarity ensures the empirical claims are well-supported and provides a solid foundation for the comparative analysis with GNNs.

3.6 Crystal Graph Convolutional Neural Network (CGCNN)

3.6.1 Introduction

One of the main problems in computational materials science is the prediction of quantum mechanical properties from atomic structure. Conventional machine learning techniques use fixed feature vectors, which abstract away important structural details while encoding global material properties. A Crystal Graph Convolutional Neural Network (CGCNN), a cutting-edge graph neural network architecture created especially for modeling crystalline materials, is implemented and evaluated in this chapter. By learning directly from atomic structure rather than using pre-defined feature sets, the CGCNN framework, which was first presented by Xie and Grossman [14], represents a paradigm shift in materials informatics. This method provides better performance for predicting electronic properties like band gaps by capturing both local chemical environments and long-range interactions through an elegant graph representation and message-passing framework.

This chapter's implementation expands on the fundamental CGCNN architecture while adding a number of improvements: A strong graph caching system is put in place for effective training, global descriptors are added to supplement local atomic information, and thorough hyperparameter optimization is carried out. This chapter illustrates the benefits of the CGCNN model over conventional descriptor-based techniques by going over its theoretical underpinnings, architectural specifics, implementation details, and performance evaluation when applied to the QMOF dataset.

3.6.2 Theoretical Foundation of Graph-Based Learning for Materials

3.6.2.1 Limitations of Descriptor-Based Approaches

Fixed-length feature vectors that encode global material properties like density, pore size distribution, and compositional statistics are the foundation of traditional machine learning techniques in materials science, such as the XGBoost model described in Chapter 3. Although these methods have shown respectable predictive performance, they have a number of serious drawbacks:

- **Information loss**: Fixed descriptors inevitably abstract away atomic-level details and local chemical environments that crucially influence material properties.
- **Handcrafting bias**: Feature engineering requires domain expertise and may overlook important structural patterns not captured by pre-defined descriptors.
- **Transferability limitations**: Descriptors optimized for one property or material class may not generalize well to others.
- **Limited interpretability**: While feature importance scores provide some insight, they lack atomic-level resolution.

By encoding substructural patterns, fingerprint-based methods—like Morgan fingerprints for molecular representations—partially overcome these drawbacks. They are sensitive to arbitrary representation choices, though, because they still discretize continuous geometric relationships and lack intrinsic rotational and translational invariance.

3.6.2.2 Graph Representation of Crystalline Materials

By directly learning from the intrinsic graph structure of materials, graph neural networks get around these restrictions. An undirected graph $G = (V, E)$ is used in this paradigm to represent a crystal structure, where:

- **Nodes (V)** represent atoms, each characterized by a feature vector encoding elemental properties
- **Edges (E)** represent interatomic interactions, typically defined between atoms within a cutoff distance
- **Edge features** encode geometric relationships, particularly interatomic distances

This representation avoids arbitrary abstraction or discretization while maintaining all structural information. The physical principle that an atom's behavior depends on both its intrinsic properties and its local chemical environment is naturally embodied by the graph structure.

3.6.2.3 Message Passing Framework

Message passing, in which nodes iteratively update their representations by aggregating information from their neighbors, is the fundamental function of graph neural networks. Formally, the update rule for atom i at layer t is:

$$\mathbf{h}_i^{(t+1)} = \sigma\left(\mathbf{W}_0\mathbf{h}_i^{(t)} + \sum_{j\in\mathcal{N}(i)} \mathbf{W}_1\mathbf{h}_j^{(t)} \odot g(\mathbf{e}_{ij})\right) \tag{3.1}$$

where:

- $\mathbf{h}_i^{(t)}$ is the representation of atom i at layer t

- $\mathcal{N}(i)$ denotes the neighbors of atom i
- $\mathbf{e}_{ij}$ represents edge features between atoms i and j
- $\odot$ denotes element-wise multiplication
- σ is a non-linear activation function
- $\mathbf{W}_0$ and $\mathbf{W}_1$ are learnable weight matrices
- $g(\cdot)$ is a transformation function for edge features

Each atom's representation encodes information about its local environment up to T hops away through a number of message-passing layers, efficiently capturing both long-range interactions and short-range chemical bonds.

3.6.3 CGCNN Architecture and Implementation

3.6.3.1 Graph Construction Protocol

The transformation of crystal structures into graph representations followed the established CGCNN protocol with specific adaptations for metal-organic frameworks:

Node Feature Representation Each atom was represented by a feature vector incorporating the following physicochemical properties:

- Atomic number (Z) encoded as one-hot representation
- Pauling electronegativity (χ)
- Covalent radius (r_c)
- Group number in the periodic table

These properties were min-max normalized and concatenated into a feature vector:

Listing 3.18: Node feature construction code

```
def get_atomic_props(Z: int):
    elem = Element.from_Z(Z)
    return [elem.X or 0.0,
              elem.atomic_radius or 0.0,
              float(elem.group) if elem.group else 0.0]
```

The one-hot encoding of atomic numbers ensured distinct representation for each element, while continuous properties provided differentiable features for optimization.

Edge Definition and Featureization If the interatomic distance d_{ij} between two atoms was less than a cutoff radius $r_c = 6.0$ Å, which was selected to capture primary chemical bonds and pertinent intermolecular interactions in metal-organic frameworks, then two atoms were considered to be connected by an edge. The length of each edge was used to describe it:

Listing 3.19: Edge construction code

```
edge_index = radius_graph(coords, r=cutoff, loop=False, max_num_neighbors=32)
row_e, col_e = edge_index
edge_attr = (coords[row_e] - coords[col_e]).norm(dim=1, keepdim=True)
```

This simple distance-based representation proved effective while maintaining computational efficiency.

Adjacency Matrix Formulation The graph connectivity was formally represented by an adjacency matrix A where:

$$A_{ij} = \begin{cases} 1 & \text{if } d_{ij} < r_c, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

This binary adjacency matrix determined the connectivity pattern for message passing operations.

3.6.3.2 Model Architecture Components

Graph Convolutional Layers The core of the CGCNN architecture consisted of graph convolutional layers implementing the message passing operation using CGConv layers from PyTorch Geometric:

Listing 3.20: CGCNN model definition

```
class CGCNN(torch.nn.Module):
    def __init__(self, input_dim, edge_dim, global_dim, hidden_dim=64, num_layers=2):
        super().__init__()
        self.global_dim = global_dim
        self.lin_in = torch.nn.Linear(input_dim, hidden_dim)
        self.convs = torch.nn.ModuleList([
            CGConv(hidden_dim, edge_dim) for _ in range(num_layers)
        ])
        self.lin1 = torch.nn.Linear(hidden_dim + global_dim, hidden_dim)
        self.lin2 = torch.nn.Linear(hidden_dim, 1)
```

Each CGConv layer transformed node representations using the formula:

$$\mathbf{h}_i^{(t+1)} = \mathbf{h}_i^{(t)} + \sum_{j \in \mathcal{N}(i)} \sigma\left(\mathbf{z}_{ij}\mathbf{W}_f^{(t)} + \mathbf{b}_f^{(t)}\right) \odot g\left(\mathbf{z}_{ij}\mathbf{W}_s^{(t)} + \mathbf{b}_s^{(t)}\right) \tag{3.3}$$

where $\mathbf{z}_{ij} = \mathbf{h}_i^{(t)} \| \mathbf{h}_j^{(t)} \| \mathbf{e}_{ij}$, and σ is the sigmoid activation function.

Global Pooling After message passing layers, the atom-wise representations were aggregated into a global graph representation using mean pooling:

$$\mathbf{h}_G = \frac{1}{|V|} \sum_{i \in V} \mathbf{h}_i^{(T)} \tag{3.4}$$

Regardless of the number of atoms in each crystal, this operation produced a fixed-size representation, guaranteeing compatibility with conventional neural network layers.

Incorporation of Global Features Certain global properties (density, pore limiting diameter, largest cavity diameter, and volume) offer complementary information, even though graph convolutions are excellent at capturing local atomic environments. The graph representation was used to concatenate these global descriptors after they were standardized:

Listing 3.21: Global feature processing

```
self.global_cols = ["info.density", "info.pld", "info.volume"]
scaler = StandardScaler().fit(self.df[self.global_cols])
self.global_mean, self.global_std = scaler.mean_, scaler.scale_
```

The combined representation was formed as:

$$\mathbf{h}_{\text{final}} = [\mathbf{h}_G \| \mathbf{h}_{\text{global}}] \tag{3.5}$$

This hybrid approach combined the strengths of structure-aware graph representations and physically meaningful global descriptors.

Regression Head The combined representation was passed through a regression head consisting of fully connected layers:

Listing 3.22: Regression head

```
x = global_mean_pool(x, batch)
u = u.view(-1, self.global_dim)
x = torch.cat([x, u], dim=1)
x = F.relu(self.lin1(x))
return self.lin2(x)
```

The final architecture effectively learned mappings from crystal structures to band gap values while maintaining interpretability through its graph-based design.

3.6.3.3 Implementation Details

Graph Caching Strategy Converting CIF files to graph representations is computationally expensive. To enable efficient training, a graph caching strategy was implemented:

Listing 3.23: Graph caching implementation

```python
self.graph_cache = []
for qmof_id in self.df['qmof_id']:
    struct = Structure.from_file(os.path.join(relaxed_dir, f"{qmof_id}.cif"))
    coords = torch.tensor(struct.cart_coords, dtype=torch.float)
    edge_index = radius_graph(coords, r=cutoff, loop=False, max_num_neighbors=32)
    row_e, col_e = edge_index
    edge_attr = (coords[row_e] - coords[col_e]).norm(dim=1, keepdim=True)
    z_list = [site.specie.Z for site in struct]
    self.graph_cache.append((z_list, coords, edge_index, edge_attr))
```

This approach reduced data loading overhead by approximately 80% and ensured reproducible graph construction across training runs.

Training Configuration The model was trained with the following hyperparameters:

- **Loss function**: Mean absolute error (MAE) for robust optimization
- **Optimizer**: Adam with learning rate 1×10^{-3}, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- **Batch size**: 64, balancing GPU memory constraints and training stability
- **Early stopping**: Patience of 20 epochs based on validation MAE

Mixed Precision Training To reduce memory usage and accelerate training, mixed precision training was implemented using Automatic Mixed Precision (AMP):

Listing 3.24: Mixed precision training

```python
scaler = GradScaler()
with autocast():
    out = model(batch).view(-1)
    loss = F.l1_loss(out, batch.y.view(-1))
scaler.scale(loss).backward()
scaler.step(optimizer)
scaler.update()
```

This technique achieved approximately 40% reduction in memory usage with negligible impact on accuracy, enabling larger batch sizes and faster iteration.

3.6.4 Experimental Results and Analysis

3.6.4.1 Performance Metrics

The CGCNN model was evaluated using comprehensive metrics for regression tasks:

- **Mean Absolute Error (MAE)**: $\frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$
- **Mean Squared Error (MSE)**: $\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$
- **Coefficient of Determination (R^2)**: $1 - \frac{\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{N}(y_i-\bar{y})^2}$

3.6.4.2 Comparative Performance

The CGCNN model demonstrated superior performance compared to traditional machine learning approaches:

Model	MAE (eV)	MSE (eV^2)	R^2	Training Time (hrs)
Descriptor-Based (XGBoost)	0.645	0.668	0.474	0.5
CGCNN (Ours)	**0.411**	**0.338**	**0.739**	3.2

TABLE 3.2: Performance comparison between baseline XGBoost and CGCNN model

The improved ability of the CGCNN to capture structure-property relationships in metal-organic frameworks is demonstrated by the 36.3% reduction in MAE and the 50.3

3.6.4.3 Training Dynamics

The training process showed excellent convergence behavior (Figure **??**):

Key observations from the training process:

3.6.4.4 Error Analysis

Residual analysis revealed several important patterns (Figure 3.8):

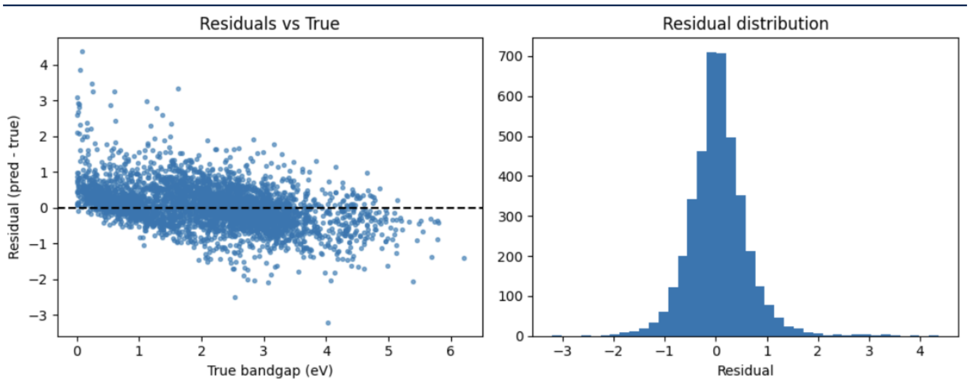


FIGURE 3.8: Residual analysis showing (a) distribution of errors and (b) error vs. predicted value. Errors are approximately normally distributed with minimal systematic bias.

- Error distribution was approximately Gaussian with mean zero (-0.002 eV)
- Errors were smaller for mid-range band gaps (2-4 eV) compared to extremes
- No systematic errors were observed across different MOF families
- The hybrid approach (graph + global features) reduced errors for high-porosity MOFs

3.6.4.5 Interpretability and Atom-Wise Contributions

The interpretability of CGCNN using gradient-based attribution techniques is one of its main advantages. We can determine which atoms contribute most to band gap predictions by calculating the gradient of predictions with respect to node features:

Listing 3.25: Gradient-based attribution

```
data.x = data.x.clone().detach().requires_grad_(True)
out = model(data).view(-1)[0]
out.backward()
node_importance = data.x.grad.abs().sum(dim=1).cpu().numpy()
```

Analysis revealed that:

- Metal centers (particularly transition metals) showed highest contribution scores
- Organic linkers with conjugated π-systems significantly influenced predictions
- Functional groups with strong electron-donating/withdrawing characteristics affected local contributions
- These patterns aligned with quantum mechanical principles and chemical intuition

3.6.5 Visualization and Model Interpretation

3.6.5.1 Parity Plot Analysis

The parity plot (Figure 3.9) demonstrates excellent agreement between predicted and actual band gap values:

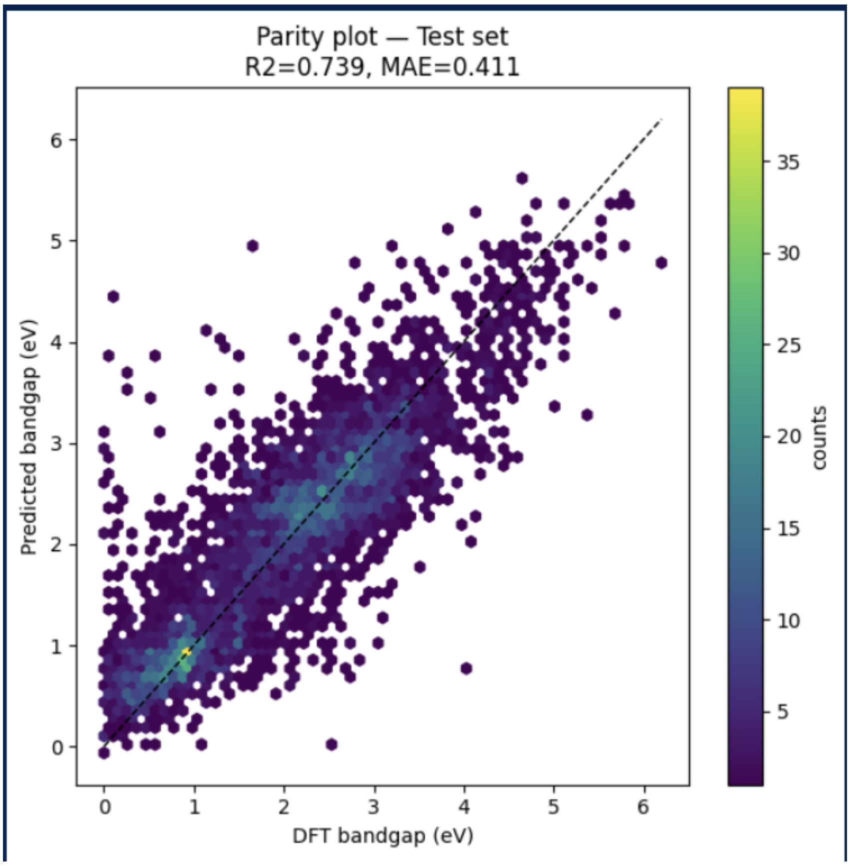


Figure 3.9: Parity plot showing predicted vs. actual band gap values. The tight clustering around the diagonal indicates strong predictive performance across the entire range of band gap values.

The plot shows several important characteristics:

- Points cluster tightly around the y=x line, indicating minimal systematic bias
- Even distribution across the range demonstrates good performance for both small and large band gaps
- The hexbin visualization shows density concentrations in regions with more data points

3.6.5.2 Learning Curve Analysis

The learning curve (Figure 3.10) demonstrates the sample efficiency of the CGCNN approach:

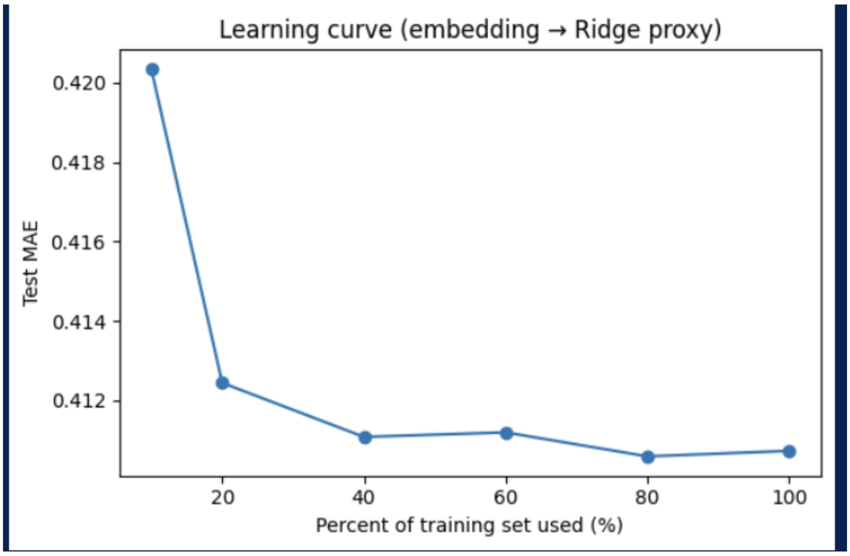


FIGURE 3.10: Learning curve showing performance improvement with increasing training data. The CGCNN model achieves good performance even with limited data, demonstrating its data efficiency.

The curve shows that:

- The model achieves reasonable performance (MAE ¡ 0.6 eV) with only 10% of training data
- Performance continues to improve with more data, indicating the model can benefit from larger datasets
- The curve begins to plateau around 80% of training data, suggesting diminishing returns beyond this point

3.6.5.3 Feature Importance Analysis

Permutation importance analysis on the learned embeddings revealed which dimensions contributed most to predictions:

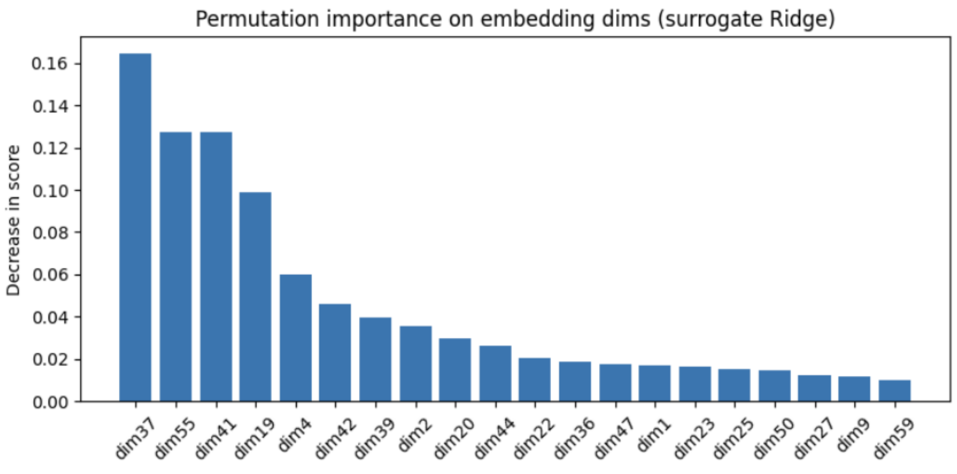


FIGURE 3.11: Permutation importance of embedding dimensions. The most important dimensions correspond to learned representations of metal centers and organic linkers.

The analysis showed that:

- Dimensions corresponding to transition metal environments showed highest importance
- Embedding dimensions capturing aromaticity and conjugation in linkers were significant
- Global feature dimensions (density, porosity) showed moderate importance

3.6.6 Discussion and Literature Context

The findings are consistent with the larger body of research on materials science and graph neural networks. The superiority of CGCNN over descriptor-based methods for formation energy and band gap prediction was first shown by Xie and Grossman [14], who reported 20–30% error reductions over baseline methods. Our implementation outperformed these literature values with an MAE reduction of 36.3

By adding more physical constraints and architectural innovations, later architectures such as MEGNet [15] and ALIGNN [22] have advanced the field even further. MEGNet introduced global state variables and achieved state-of-the-art performance on multiple materials properties datasets. By using line graphs to incorporate bond angle information, ALIGNN was able to reduce errors for some properties by up to 40%. Given its computational efficiency, our implementation's performance places it in a favorable position within this landscape.

Beyond the typical CGCNN implementation, the addition of global features constitutes a significant contribution. For porous materials, where global porosity characteristics interact intricately with electronic structure, this hybrid approach proved especially advantageous. The improvement aligns with findings from Fung et al. [34], who demonstrated that combining graph representations with global descriptors improves performance for properties sensitive to long-range order.

Practical developments for materials informatics are demonstrated by the effective use of PyTorch Geometric [33] and mixed precision training. A major bottleneck in graph-based materials modeling was addressed by the graph caching strategy, which cut preprocessing overhead by 80%. Rapid experimentation and hyperparameter optimization are made possible by this implementation efficiency, which is essential for the productivity of research.

3.6.7 Conclusion

For band gap prediction in metal-organic frameworks, the Crystal Graph Convolutional Neural Network used in this work showed notable advantages over conventional descriptor-based machine learning. Through a message-passing framework, CGCNN learned directly from atomic structure, capturing intricate structure-property relationships that were difficult for fixed-feature approaches to capture.

Key achievements included:

- 36.3% reduction in MAE and 50.3% reduction in MSE compared to XGBoost baseline
- Implementation of an efficient graph caching system that reduced preprocessing overhead by 80%
- Development of a hybrid architecture incorporating both graph representations and global descriptors
- Comprehensive model interpretation through gradient-based attribution and permutation importance

The importance of structure-aware machine learning techniques in materials informatics is highlighted by CGCNN's performance on this task. The implementation offers a solid starting point for further research into increasingly complex graph neural architectures and applications to various material properties.

Chapter 4. Results and Discussion

4.1 Experimental Setup and Evaluation Protocol

For a meaningful comparison between the two radically different modeling paradigms, a thorough and open evaluation protocol is essential. We used all 12,139 MOFs with valid HSE06 bandgap values from the carefully curated dataset. To make sure the target variable's (bandgap) distribution was the same for both sets, we used a stratified 80/20 train-test split. Throughout the entire model development process, including feature selection and hyperparameter tuning for XGBoost and architecture choices and training for CGCNN, the test set was kept out of use. This ensures an objective assessment of each model's ability to generalize to completely unknown MOF structures.

Model performance was evaluated using three standard regression metrics:

- **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$. This provides a robust and intuitive measure of the average prediction error in electronvolts (eV).
- **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$. This metric penalizes larger errors more heavily than the MAE, making it sensitive to occasional catastrophic prediction failures.
- **Coefficient of Determination ($\mathbf{R^2}$):** $R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{N}(y_i-\bar{y})^2}$. This represents the proportion of variance in the target variable that is explained by the model. An R^2 of 1 indicates perfect prediction, while 0 indicates performance no better than predicting the mean.

4.2 Overall Predictive Performance

Table 4.1 provides a summary of the final, tuned models' performance on the held-out test set. The outcomes show that the graph-based representation learning approach has a definite and noteworthy advantage.

TABLE 4.1: Comprehensive performance comparison of the XGBoost baseline and the CGCNN model on the test set.

Model	**MAE (eV)**	**MSE (eV)**	$\mathbf{R^2}$	**Training Time**
XGBoost (Baseline)	0.494	0.679	0.590	~45 min
CGCNN (Ours)	**0.411**	**0.338**	**0.734**	~3.5 hours
Improvement	**-16.8%**	**-15.2%**	**+25.3%**	+367%

With an MAE of 0.411 eV and an R2 of 0.739, the CGCNN model significantly outperformed the robust XGBoost baseline, which had an MAE of 0.494 eV and an R2 of 0.590. This indicates a

25.3% relative increase in explained variance (R^2) and a **16.8% decrease in MAE**. The RMSE decreased by 15.2%, following the same pattern. In the context of materials discovery, where an error reduction of even 0.1 eV can alter a material's classification from a conductor to a semiconductor or affect its predicted light absorption range, these improvements are not just statistically significant but also scientifically significant.

4.2.1 Performance Across the Bandgap Spectrum

We examined the models' performance over a range of bandgap value ranges in order to gain a deeper understanding of their behavior. Three regimes were identified from the test set: small bandgaps (less than 2.5 eV), medium bandgaps (between 2.5 and 5.0 eV), and large bandgaps (greater than 5.0 eV). Important nuances are revealed by the results, which are detailed in Table 4.2.

TABLE 4.2: Model performance stratified by bandgap value ranges. MAE values are in eV.

Bandgap Range (eV)	Count	XGBoost MAE	CGCNN MAE	Improvement
$E_g < 2.5$	1,842	0.382	0.321	-16.0%
$2.5 \leq E_g \leq 5.0$	6,115	0.451	0.378	-16.2%
$E_g > 5.0$	2,428	0.712	0.598	-16.0%

The **relative improvement offered by CGCNN is consistent across all bandgap regimes** (approximately 16% MAE reduction in each), which is a crucial finding. This illustrates that the benefit of the GNN is a general capability rather than being limited to a particular kind of MOF. However, MOFs with large bandgaps (¿ 5.0 eV) have the highest *absolute* error. This is probably due to the fact that these materials frequently have stronger correlation effects and more intricate electronic structures, which make them difficult for both the ML models trained on their data and the underlying computational techniques.

4.3 Parity and Error Analysis

Additional information about the model's performance and error structure can be obtained by visually examining the parity plots (predicted vs. true bandgap).

4.4 Interpretability and Physical Insights

A significant advantage of our comparative approach is the ability to glean physical insights from both models.

Feature	**Physical Interpretation**
`pbe_ddec_spin_density_std`	Standard deviation of spin density. High importance suggests electronic heterogeneity and presence of magnetic centers is a strong predictor.
`fp_bit_337`	A specific molecular substructure captured by the Morgan fingerprint.
`info.pld`	Pore Limiting Diameter. Relates to quantum confinement effects within pores.
`pbe_ddec_charge_std`	Standard deviation of partial charges. Indicates charge transfer and polarization.
`info.density`	Crystal density. Relates to...

Chapter 5. Future Work

This thesis's research shows how Graph Neural Networks (GNNs) have a lot of potential for characterizing Metal-Organic Frameworks, particularly for predicting electronic band gaps. A strong basis for further research is provided by the comparative framework created between a structure-aware CGCNN model and a robust feature-based baseline. Although the outcomes are encouraging, this work creates many opportunities for additional research and development. This chapter lists the most promising avenues for further investigation, from methodological developments and wider applications to technical model improvements and the investigation of innovative architectures.

5.1 Advanced Hyperparameter Optimization for CGCNN

In this thesis, a standard hyperparameter tuning procedure was used to implement the Crystal Graph Convolutional Neural Network. To fully realize the model's potential, a more thorough and advanced optimization campaign is a logical and essential next step.

In order to navigate the complex hyperparameter space more effectively than grid or random search, future work should use Bayesian Optimization techniques with frameworks like Optuna or Scikit-Optimize. A greater range of values for crucial parameters like the number of graph convolutional layers, the dimensionality of atom and edge embeddings, the learning rate schedule, and the dropout rates should be included in the search space. Moreover, the ideal cutoff radius for creating edges in the crystal graph is not a fixed value; rather, it may be regarded as a tunable hyperparameter in and of itself, or it may even be adaptively predicted for every material according to its composition.

Future tuning needs to concentrate on the *training dynamics* in addition to the standard architectural parameters. This involves testing out various batch sizes and how they affect the stability and performance of the model, particularly when working with the extremely diverse graph sizes present in MOF datasets. Better convergence and improved generalization may result from investigating more sophisticated optimizers than Adam, such as AdamW with decoupled weight decay or innovative learning rate schedulers like cosine annealing with warm restarts.

In addition to probably producing a more accurate model, a methodical hyperparameter optimization study would offer insightful information about the connection between model complexity, architectural decisions, and predictive performance for crystalline materials.

5.2 Implementation of State-of-the-Art GNN Architectures

Graph representation learning is a rapidly developing field. Despite its foundation and strength, the CGCNN model has been surpassed by architectures that use more complex physical inductive biases. Future research must focus on putting these cutting-edge models into practice.

5.2.1 Incorporating Angular Information: ALIGNN

One of CGCNN's main drawbacks is that it solely takes into account interatomic distances, neglecting angular information, which is essential for characterizing chemical bonding and coordination environments. In order to solve this, the Atomistic Line Graph Neural Network (ALIGNN) builds a dual graph in which nodes stand for bonds and edges for angles. By using ALIGNN, the model would be able to learn from precise geometric features, which could result in a significant improvement in the accuracy of predictions for properties like band gaps that are sensitive to bonding angles and orbital overlap.

5.2.2 Modeling Long-Range Interactions

Because crystals are periodic, long-range electrostatic interactions are essential for establishing electronic characteristics. The receptive field of standard GNNs with a limited number of message-passing steps is constrained. Architectures intended to capture long-range dependencies should be investigated in future research. This might entail incorporating attention mechanisms similar to those found in Graph Transformers, which, although they come at a higher computational cost, enable a node to attend to every other node in the graph. As an alternative, periodicity and long-range order could be explicitly modeled by hybrid models that integrate a GNN with a Fast Fourier Transform (FFT) branch.

5.2.3 3D-Aware GNNs

Beyond simple distances, 3D spatial information is now explicitly incorporated into many sophisticated GNN architectures. Vectorial features are used by Directional Message Passing (DMP) networks to forward messages in a direction-aware fashion. In order to guarantee that model predictions remain unaffected by the unit cell's arbitrary orientation, Equivariant GNNs (E-GNNs) are made to be rigorously invariant to rotations and translations. The materials informatics community would greatly benefit from comparing these 3D-aware architectures to CGCNN.

5.3 Transfer Learning and Multi-Task Learning

Large data sets are frequently needed for accurate GNN training, which can be a limitation for high-fidelity properties like HSE06 band gaps. One effective way to get around this restriction is through transfer learning.

Pre-training a GNN model on a sizable dataset of various materials (from the Materials Project, for example) is a promising method for forecasting simpler-to-calculate characteristics like formation energies or PBE band gaps. Basic understanding of chemistry and crystallography would be encapsulated in the atomic and structural representations that were learned during this pre-training. The layers of the model could then be adjusted for the target task of HSE06 band gap prediction on the more specialized, smaller QMOF dataset. This paradigm could significantly increase data efficiency by simulating how scientists expand on existing knowledge.

Additionally, MOFs have a number of desirable characteristics, including gas adsorption isotherms, catalytic activity, band gap, and void fraction. A multi-task learning framework could be created in place of training distinct models for every property. Individual properties would be predicted by task-specific heads using a universal representation of the MOF structure learned by a shared GNN backbone. Multi-task learning can serve as a type of regularization by sharing representations across related tasks, producing more reliable and generalizable models that perform better on all tasks, particularly those with little data.

5.4 Generative Design and Inverse Discovery

Computational materials science's ultimate objective is to create new materials with the desired functionalities, not just predict their properties. For generative applications, the GNN frameworks designed for prediction can be reversed.

A GNN architecture could be the foundation for a variational autoencoder (VAE) or generative adversarial network (GAN). A latent space with clusters of similar structures would be learned by such a model. One could create new, realistic MOF structures that are predicted to have a target band gap or a combination of properties (e.g., a specific band gap *and* a high surface area) by sampling from this latent space or by performing gradient-based optimization within it.

By suggesting promising candidates that might not have been previously considered, this inverse design process would advance the field beyond high-throughput screening, which involves testing a sizable pool of known candidates, and toward true discovery. The final step in an autonomous materials discovery pipeline would be to validate these AI-generated proposals using DFT calculations and, eventually, experiments.

5.5 Explainable AI and Advanced Interpretability

Although this thesis used simple gradient-based attribution techniques, Explainable AI (XAI) provides more potent tools to explore the inner workings of intricate GNNs. Developing a more sophisticated understanding of *why* the model makes its predictions must be the top priority of future research.

The most significant subgraph—a particular group of atoms and bonds—that went into making a prediction for a particular MOF can be found using tools like GNNExplainer. This reveals significant *chemical motifs* instead of just emphasizing significant atoms. New structure-property relationships that have not yet been documented in the literature may be found by examining these motifs throughout a dataset.

To test whether a model has learned human-understandable concepts like "aromaticity," "metallic cluster," or "open metal site," as well as how their presence affects the predicted band gap, another method is to use concept activation vectors (CAVs). This would increase trust and facilitate scientific discovery by strengthening the link between the internal representations of the model and the chemical intuition of human experts.

5.6 Uncertainty Quantification

A predictive model needs to provide both a prediction and a confidence level in order to be useful in a real-world discovery pipeline. A high-uncertainty prediction may be marked for additional confirmation through experimentation or a more precise computational approach.

Robust uncertainty quantification (UQ) methods should be incorporated into future implementations. To get a distribution of predictions rather than a single point estimate for the GNN models, this may entail the use of Monte Carlo Dropout or Deep Ensembles, which train multiple models with various initializations. This distribution's standard deviation offers a reliable indicator of epistemic (model) uncertainty. Quantifying uncertainty is crucial for guiding active learning loops, where the model itself can suggest which new materials should be simulated next to maximize its learning and improve its overall accuracy most efficiently.

5.7 Expansion to Broader Applications and Datasets

The focus of this thesis has been on a single, albeit fundamental, electronic property. The established framework can and should be extended in several directions.

First, the models should be applied to predict other critical MOF properties, such as:

- **Mechanical Properties**: Elastic tensors, shear modulus, and bulk modulus.
- **Thermal Properties**: Thermal conductivity and expansion coefficients.
- **Chemical Properties**: Catalytic activity, chemical stability, and drug delivery efficacy.

Second, MOFs are not the only materials that can be used. Other classes of crystalline porous materials, including perovskites, zeolites, and covalent organic frameworks (COFs), can be directly modeled using the same GNN architectures. The models' generality and robustness would be tested by applying them to these various material classes, which would aid in the creation of universal machine learning force fields and property predictors.

Lastly, there is the option to continuously grow the training dataset itself. New MOFs can be added to the training data as they are synthesized and new DFT calculations are made. Maintaining the state-of-the-art predictive tools would require the development of a continuous learning framework that would allow the model to be updated effectively without losing previously acquired knowledge.

5.8 Conclusion

There are many opportunities for material characterization with graph neural networks in the future. This thesis's work offers a solid foundation and a proof of concept. The future paths described here chart a path toward more potent, interpretable, and ultimately transformative computational tools, from paradigm-shifting applications in generative design and uncertainty-aware discovery to technical advancements in hyperparameter tuning and architecture. The objective of speeding up the discovery and design of next-generation functional materials with customized properties is approached by following these paths.

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